VCU220

VCU220 CONCEPT, RA, FSC & PROGRAM PLANNING

Centralized Vehicle Control Unit & Infotainment Domain Controller - 220 kW Peak EV

Document ID	VCU-EV220-DOC1-CONCEPT-PLAN-001
Revision	Rev A2 - Final Technical Hardening
Technical Freeze / Authority	JAN-2026 Program Baseline / Rev A2 hardening review 14-Aug-2026
Vehicle	220 kW peak / 110 kW continuous EV platform
Architecture	Central ECU with independent safety MCU + high-performance cockpit/application SoC
Process Scope	Automotive SPICE 4.0 SYS.1 / SYS.2 / SYS.3 with HWE.1 / SWE.1 / SYS.4 / SYS.5 / VAL.1 handoff
Safety / Cybersecurity	ISO 26262 vehicle safety lifecycle; ISO/SAE 21434 TARA; final ASIL from vehicle HARA/FSC/TSC
Evidence Status	Engineering baseline. Planned requirements/tests are not executed release or compliance evidence.

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1. Executive Purpose and System Mission

This document establishes the reusable concept, requirements-authority, architecture, technology-selection and program-planning baseline for a centralized VCU/cockpit domain on a 220 kW peak electric-vehicle platform. The architecture intentionally consolidates vehicle services and cockpit functions while retaining an independent safety-control path for authoritative propulsion, high-voltage and critical fallback decisions.

Rev A1 architecture hardening: Centralization is defined as one central ECU/housing, not one undifferentiated software domain. The independent safety MCU remains the authoritative torque/HV controller; the HPA SoC and Android/POSIX domains are treated as non-authoritative unless a future safety decomposition is separately justified.

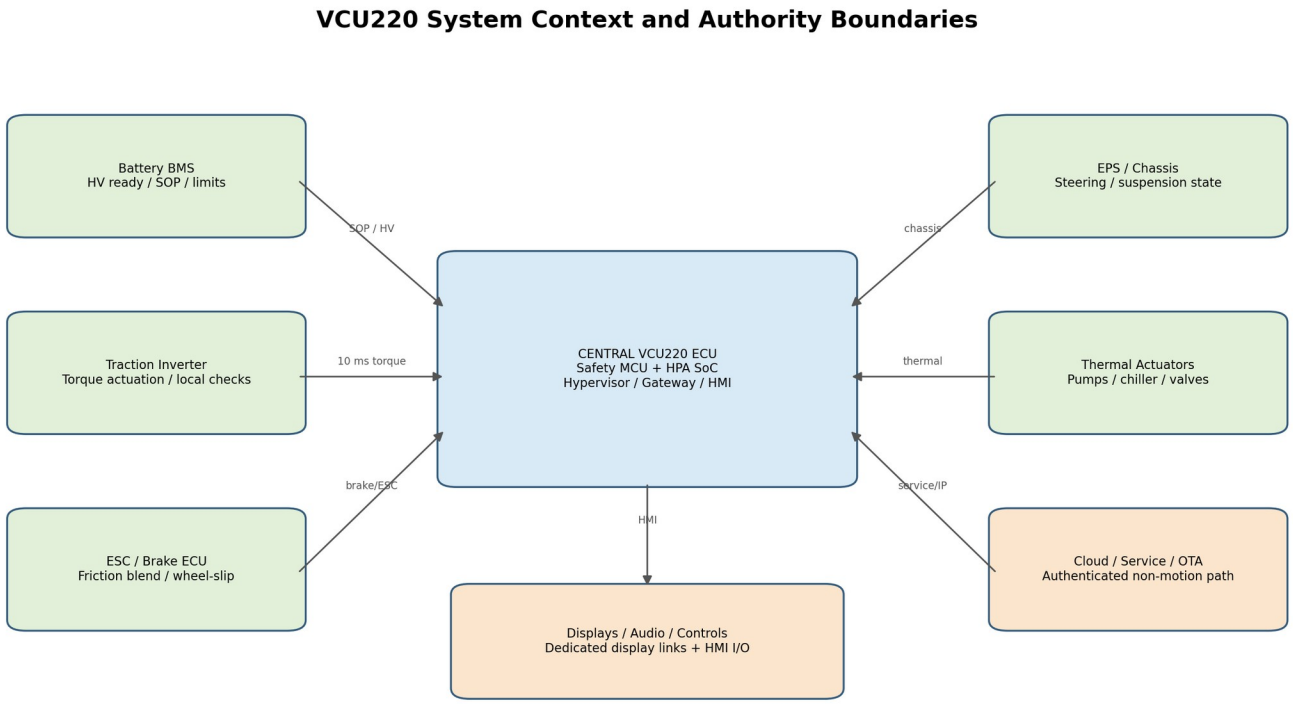


Figure 1 - System context and external authority boundaries

1.1 Design objectives

- Consolidate legacy VCU, gateway and cockpit services without transferring unrestricted motion authority into QM infotainment software.
- Support 220 kW peak and 110 kW continuous propulsion capability through bounded torque/regen/HV/thermal coordination rather than direct inverter switching control.
- Provide a future-ready compute and network architecture that can scale from 400 V to 800 V vehicle variants and from 1 Gbit/s vehicle IP to higher-rate zonal/central backbones.
- Preserve deterministic safety operation under HPA SoC reset, Android crash, hypervisor fault, communication loss, low-voltage disturbance and thermal derating.
- Maintain bidirectional traceability from stakeholder need through SYS.3 architecture, safety allocation, DVP and vehicle validation.

2. Controlled Technical Baseline

Parameter	Rev A1 Baseline	Qualification
Technical freeze	JAN-2026	Later standards/product changes require change-impact review.
Vehicle propulsion	220 kW peak / 110 kW continuous	Actual motor/inverter torque-speed and BMS SOP maps are controlled ICD inputs.
HV architecture	400 V nominal reference; 800 V scalable variant	Current ratings/calibration are variant-specific.

Parameter	Rev A1 Baseline	Qualification
Central ECU power	70 W nominal / 130 W sustained peak; 150 W short input	Excludes external displays, amplifier/speakers and powered peripherals unless reallocated.
Torque latency	<=15 ms pedal-to-torque request	Internal timing budget includes 5 ms reserve.
Safety reaction	<=10 ms internal electronic inhibit request	Vehicle hazard FTTI may differ; FSC/TSC controls final allocation.
CAN FD	500 kbit/s arbitration / 2 Mbit/s data phase	Final bus load and worst-case response evaluated by network tool.
Ethernet	1000BASE-T1 baseline	2.5/10G option reserved; pixels on dedicated display transport.
Safety allocation	SG-01 unintended/excess torque ASIL D target; remaining safety allocations per HARA	SG-01 preliminary HARA may require ASIL D.

3. Boundaries, Assumptions and Constraints

ID	Topic	Assumption	Closure Rule
A-001	HV Architecture	400 Vdc nominal traction battery; 220 kW peak / 110 kW continuous propulsion reference. An 800 V variant is supported by interface/calibration scaling rather than assuming identical current limits.	Freeze vehicle BMS/inverter voltage-current envelopes and connector ICD.
A-002	Centralization	The product is one central ECU/housing containing an independent safety MCU and a high-performance application SoC. Safety-critical torque authority does not depend on the Android/infotainment guest.	Close silicon/hypervisor safety concept and common-cause analysis.
A-003	Actuation Boundary	VCU sends bounded torque/deceleration/HV requests. Inverter, ESC/brake ECU, EPS and BMS retain local actuator-level safety checks and hardware protection.	Freeze each actuator ICD and safety manual assumptions.
A-004	Driver Modes	Eco, Normal, Sport, Track and Valet/Custom are calibration profiles; none may bypass battery/inverter/ESC/EPS thermal or safety hard limits.	Approve mode safety analysis and homologation/market rules.
A-005	Track Mode	Track mode may be market-, geofence-, service- or closed-course restricted. ESC retains the final stability envelope; VCU only requests a permitted mode.	Freeze legal/homologation and ESC feature contract.
A-006	Power Budget	70 W nominal and 130 W sustained peak with a 150 W short electrical input envelope apply to the central ECU electronics, excluding display backlights, speakers/amplifier and externally powered antennas/sensors.	Close power tree and vehicle LV budget.
A-007	Network	Safety motion commands use redundant or independent CAN FD channels; Ethernet carries service-oriented, diagnostic and cockpit traffic. Raw display pixels use dedicated display SerDes/DP links, not the 1-Gbps vehicle Ethernet baseline.	Freeze E/E architecture and traffic matrix.
A-008	Safety	Program target is ASIL C for torque arbitration; preliminary HARA may allocate ASIL D safety goals for unintended torque. Hardware/software	Complete item definition/HARA/FSC/TSC.

ID	Topic	Assumption	Closure Rule
		architecture shall not prevent ASIL D-capable mitigation if HARA requires it.	
A-009	Cockpit OS	Android Automotive/Linux/QNX guests are treated as QM unless specific functions receive a higher allocation. Safety-relevant telltales use an independent monitored path.	Freeze HMI safety concept.
A-010	Thermal Limits	Battery/inverter temperature thresholds are supplier-calibration inputs; the VCU shall not hard-code a universal 65 C protection threshold.	Freeze BMS/inverter thermal limit ICD.
A-011	Vehicle Dynamics	VCU coordinates high-level longitudinal/chassis requests; ESC/ABS retains wheel-slip and hydraulic brake authority and EPS retains steering hardware safety.	Close chassis arbitration ICD.
A-012	OTA	Vehicle supports secure OTA and workshop update with A/B or equivalent recovery. Update engineering follows ISO 24089 and applicable UNECE R156 market obligations.	Freeze SUMS/update architecture.

ID	Area	Constraint
C-001	Power	Central ECU <=45 W nominal and <=75 W peak; short transient may exceed only if vehicle LV and thermal design approve.
C-002	Control Latency	Pedal-to-valid-torque-request <=15 ms end-to-end reference; internal VCU compute/network allocation <=6 ms target.
C-003	Safety Reaction	Electronic safety monitor to propulsion inhibit/request <=10 ms internal capability target; final hazard FTTI follows FSC/TSC.
C-004	CAN FD	Powertrain safety networks: 500 kbps arbitration / 2 Mbps data baseline; nominal sustained utilization <=40%, fault burst <=60%.
C-005	Ethernet	1000BASE-T1 baseline for service/cockpit/gateway traffic; architecture shall support 2.5/10G uplink option if future central compute traffic requires it.
C-006	Thermal	130 W sustained-peak central ECU shall close Tj/case/junction margins at released ambient with >=20% thermal design margin.
C-007	Boot	Safety MCU motion-control availability <=150 ms after wake reference; cockpit fast-visible telltale <=500 ms; rich Android UX may boot later.
C-008	Security	Infotainment/telematics compromise shall not create a direct command path to torque/HV actuation.
C-009	Mode Transition	Mode request acknowledgement <=50 ms; torque-map blending is jerk-bounded and may take 200-500 ms or longer depending torque delta.
C-010	Calibration	All power/torque/regen/thermal/mode safety-relevant calibration is signed, variant-bound, range-checked and rollback controlled.
C-011	EMC	CISPR 25 and ISO 11452 component-level targets plus OEM vehicle EMC profile apply to central ECU and harness.
C-012	Electrical	ISO 16750-2:2023 and ISO 7637-2:2011 baseline for 12 V/24 V electrical loads/transients; vehicle-specific LV profile is controlling.

4. Architecture Principles and Controlled Decisions

ADR	Decision Area	Decision	Status	Rationale
ADR-001	Central compute topology	One central ECU with independent safety MCU + HPA application SoC	Selected	Maintains infotainment consolidation while keeping torque/HV authority independent from Android/Linux faults.
ADR-002	Torque execution	Safety MCU owns final torque arbitration and safety clamp; HPA SoC may compute advisory/comfort requests	Selected	Reduces hypervisor/Android qualification burden and gives deterministic WCET.
ADR-003	Cockpit SoC	R-Car X5H / Snapdragon Cockpit Elite class evaluated; no silicon frozen in concept baseline	Open RFQ	Future-proof multi-display/AI performance requires workload benchmark, supplier roadmap and power evidence.
ADR-004	Real-time central processor	AURIX TC4x primary safety-MCU reference; S32N55 class alternative for deeper real-time consolidation	Open RFQ	Both target centralized safety/real-time automotive control; final choice by timing, I/O, safety, security, ecosystem and lifecycle.
ADR-005	Hypervisor	Safety-qualified Type-1/partitioning hypervisor for HPA SoC; QNX Hypervisor for Safety class reference	Selected principle	QM Android/Linux guests must be isolated from safety/cluster services and hardware resources.
ADR-006	Driver-mode source	HMI is a request source only; safety MCU validates mode against vehicle state, market, thermal and chassis permissions	Selected	Prevents infotainment compromise or UI fault from directly selecting unsafe torque/stability behavior.
ADR-007	Track mode	ESC final authority; VCU requests Track calibration but cannot defeat ABS/ESC hardware protections	Selected	Mode excitement cannot override safety limits or local actuator protections.
ADR-008	HV contactors	VCU owns high-level HV state orchestration; BMS owns pack contactor drive and pack-level protection	Selected	Avoids duplicated pack-safety ownership and keeps pack hardware protection local.
ADR-009	Display transport	Display pixels over DP/eDP/automotive SerDes; Ethernet for services/state/updates	Selected	1 Gbps Ethernet should not be consumed by raw multi-display pixels.
ADR-010	Safety comm	CAN FD A/B or independent safety channels for torque/brake/HV; Ethernet not sole safe motion path	Selected	Preserves actuation during Ethernet switch/SoC faults.
ADR-011	E2E vs SecOC	AUTOSAR E2E protects communication integrity; SecOC separately provides authenticity/freshness	Selected	Avoids treating CRC/counter as a cryptographic signature.
ADR-012	Thermal thresholds	VCU consumes normalized BMS/inverter thermal limits/SOP rather than universal hard-coded 65 C limits	Selected	Cell/module/inverter hardware have different valid thermal envelopes.

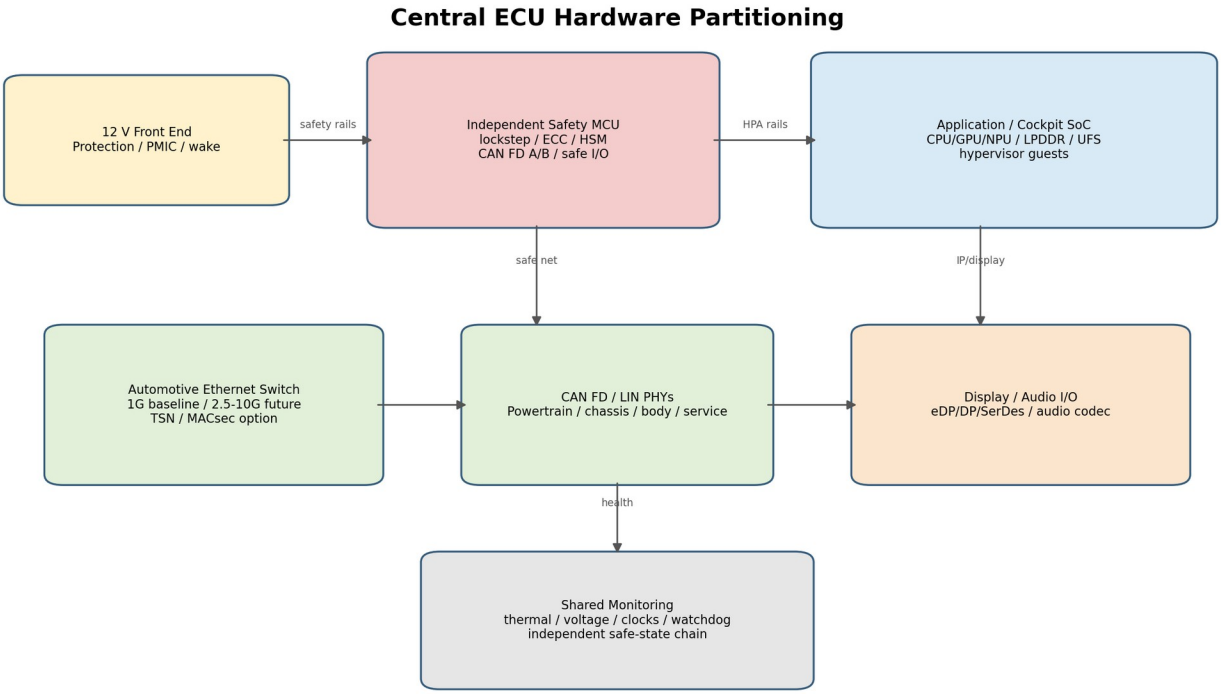


Figure 2 - Central ECU hardware partitioning

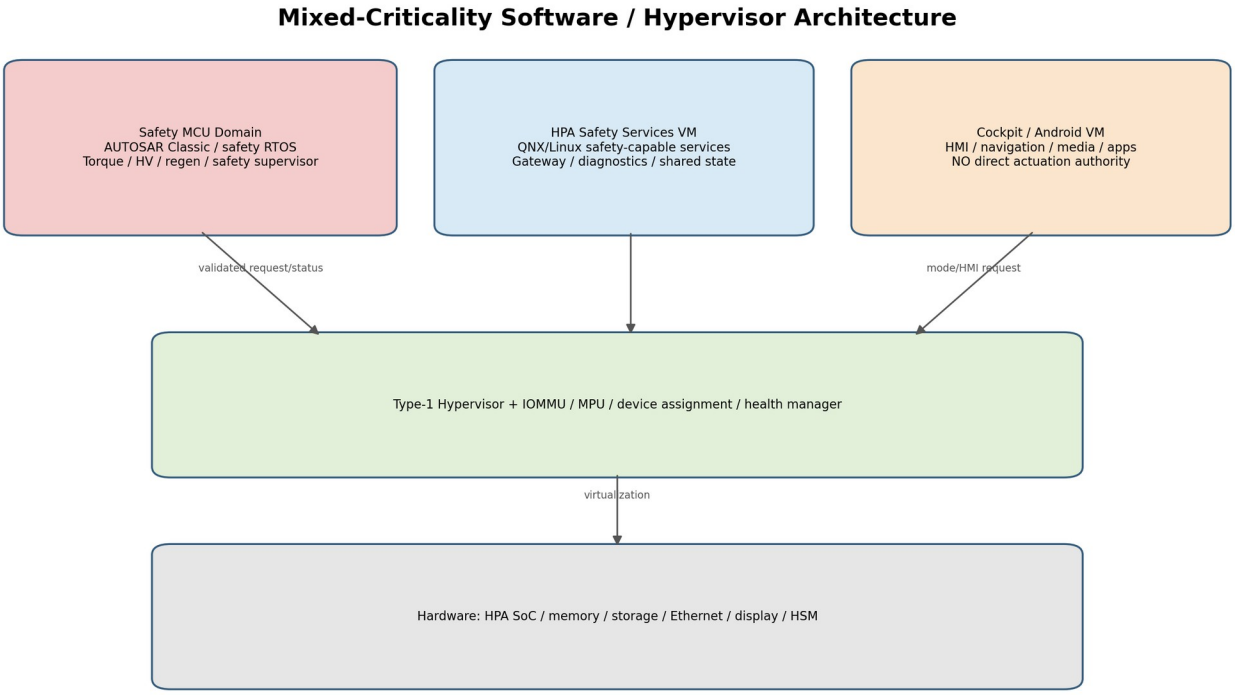


Figure 3 - Mixed-criticality software and hypervisor architecture

5. Driver Mode Management Concept

Driver-mode management is a calibration and arbitration function, not a direct actuator bypass. Each requested profile is accepted only after safety-domain validation of vehicle state, user authorization, market configuration, thermal/SOP capability and local ECU readiness. The HMI may acknowledge the accepted mode within 50 ms while the torque and regenerative-braking maps are blended separately over a jerk-bounded interval, typically 200-500 ms as a design-start range.

Mode	Pedal / Torque Feel	Power Cap	Regen	Thermal	Chassis / ESC Request	Safety Boundary
Eco	Damped progressive map	110 kW reference cap	High, subject to traction/SOC/SOP	Eco thermal/HVAC priority	Standard/strict; no relaxation	Mode cap never overrides minimum safety/thermal limits.
Normal	Linear balanced	Up to 220 kW if SOP permits	Medium / natural decel	Balanced	Standard	Default road mode.
Sport	Fast progressive	Up to 220 kW if SOP permits	Low-medium, configurable	Battery/inverter pre-conditioning request	Permitted dynamic calibration only	ESC owns final slip/yaw limits.
Track	Non-linear track map	220 kW + only supplier-approved time-limited boost	Configurable within brake/traction limits	Maximum pre-chill request	Track request subject to ESC approval	May be geofence/market/closed-course restricted.
Valet / Custom	User or fleet constrained	80 kW default valet cap; configurable lower	Standard	Standard	Maximum restriction	Signed policy; speed/torque/geofence restrictions cannot be raised without authorization.

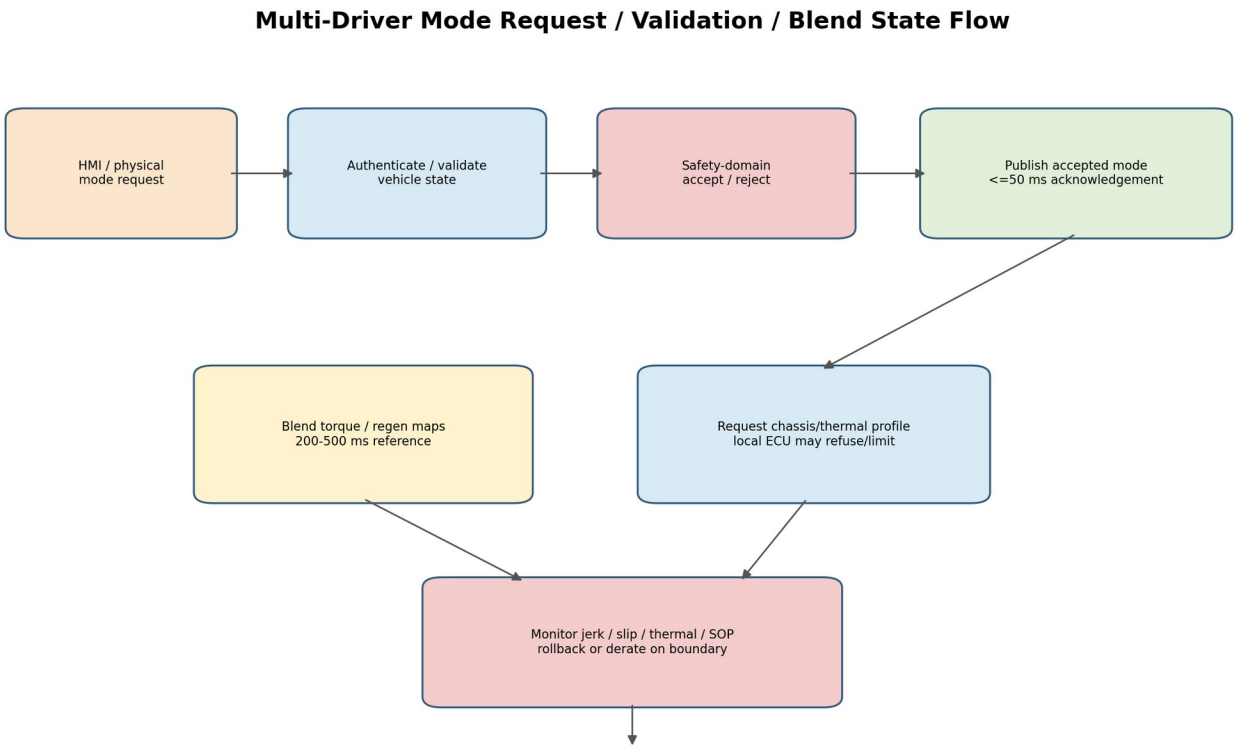


Figure 4 - Driver mode validation and transition flow

6. Network, Timing and Power Feasibility

The safety path uses CAN FD with deterministic scheduling and local actuator safety checks; Ethernet supports diagnostics, services, cockpit and future zonal integration but is not the sole path for safety torque or HV control. The reference CAN traffic estimate is intentionally conservative and is closed in the master workbook using a final CANoe/CANalyzer schedule including bit stuffing, arbitration, retransmission and fault traffic.

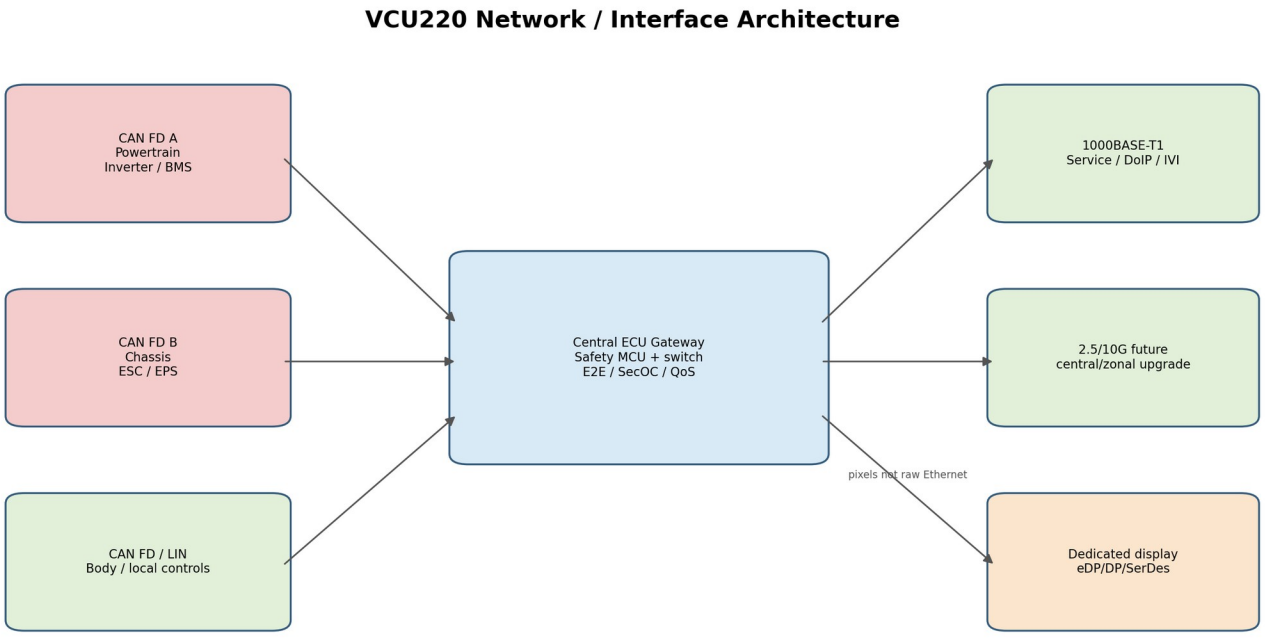


Figure 5 - Safety and IP network architecture

6.1 Pedal-to-torque latency budget

Stage	Budget ms
Pedal/input acquisition + qualification	2.0
Torque arbitration / limits / mode mapping	2.0
Safety monitor / authorization	1.0
CAN scheduling/transmit to inverter	2.0
Inverter reception/local plausibility	3.0
Vehicle-level reserve	5.0

6.2 Internal safety-reaction capability budget

Stage	Budget ms
Safety monitor cycle	1.0
Fault qualification	2.0
Safe-state decision	1.0
CAN/inhibit transmit	2.0
Inverter local inhibit/ack reference	4.0

7. Technology and Component Strategy

No single processor is frozen at concept release. The central ECU is deliberately decomposed into a deterministic safety-control processor and a high-performance application/cockpit processor so supplier, workload and lifecycle choices can evolve without redefining the vehicle motion-safety concept.

Candidate	Role / Strength	Safety / Ecosystem	Limitation	Benefit	Program Position
Infineon AURIX TC4x	High deterministic real-time MCU; 6-core TC4Dx class with lockstep and eMobility/vehicle-motion positioning	Strong safety MCU reference; AUTOSAR ecosystem	Not a premium Android/graphics cockpit SoC	Excellent independent torque/HV safety authority	Use as primary safety MCU reference
NXP S32N55	Centralized safe real-time	Very strong real-time	Premium cockpit	Could consolidate multiple	Alternative vehicle-control

Candidate	Role / Strength	Safety / Ecosystem	Limitation	Benefit	Program Position
	super-integration with hardware isolation and Ethernet/security	central controller capability	GPU/Android may still need HPA pairing	body/chassis/VCU functions	processor; evaluate Gen2 consolidation
Renesas R-Car X5H	Multi-domain HPA with 32 A-class cores, R52 lockstep safety cores, GPU/AI and multi-display	ASIL-capable safety cores plus HPA	Early-generation maturity/sampling and power/BOM risk at Jan-2026 freeze	Open Android/Linux/XEN/QNX ecosystem and central compute roadmap	Primary future-proof HPA reference, but keep external safety MCU
Qualcomm Snapdragon Cockpit Elite	Premium cockpit HPA, strong CPU/GPU/AI and hypervisor-oriented virtualized cockpit	Safety/security-focused automotive platform	Vendor ecosystem lock-in, power/BOM and low-level deterministic control not the primary VCU strength	Premium cockpit UX and future central-compute scalability	Strong HPA alternative; keep external safety MCU

7.1 Architecture BOM and Evidence Expectations

Domain	Item	Qty	Primary/Class	Alternate	Selection Criteria	Justification	Required Evidence	Status
Compute	Safety MCU	1	Infineon AURIX TC4x class	Renesas RH850/U2B / NXP S32N class	Lockstep, ECC, HSM, CAN/Ethernet, safety manual, AUTOSAR ecosystem, lifecycle	Independent deterministic authority for torque/HV supervision	Safety manual, FMEDA, MCU benchmark, PCN/EOL	RFQ
Compute	Application/ Cockpit SoC	1	Renesas R-Car X5H class	Qualcomm Snapdragon Cockpit Elite class	CPU/GPU, virtualization, multi-display, memory BW, power, safety island, Android/Linux/QNX support, lifecycle	Supports premium cockpit + centralized services while torque remains external on safety MCU	Target workload benchmark, thermal, safety/security collateral	RFQ
Compute	Safety-centric central processor alternate	1 option	NXP S32N55 class	AURIX TC4x + gateway FPGA/ASIC	Deterministic real-time consolidation, hardware isolation, Ethernet switch/security	Strong central vehicle control option if premium cockpit remains separate/paired	NXP safety/performanc e evidence	Trade study
Software	Safety hypervisor	1	QNX Hypervisor for Safety class	Green Hills INTEGRITY Multivisor class / supplier safety hypervisor	ISO26262 evidence, guest isolation, Android/Linux support, device assignment, boot time	Reduces mixed-criticality integration risk	Safety certificate/manual, latency, supported SoC	RFQ
Software	Safety OS/AUTOSAR Classic	1	AUTOSAR Classic OS/BSW + safety vendor stack	SafeRTOS / proprietary safety RTOS as project architecture permits	Timing, MCAL, safety qualification, diagnostics, SecOC/E2E	Deterministic safety MCU platform	Safety qualification/toolch ain evidence	RFQ
Software	Cockpit OS	1	Android Automotive + Linux/QNX services	QNX cockpit / AGL	OEM UX ecosystem, app model, boot time, security, lifecycle	Rich HMI kept behind safety boundary	BSP, update, security support	RFQ
Power	Safety PMIC	1	Infineon TLF35585 class	NXP automotive safety PMIC class	3-40V input, watchdog, rail monitoring, safe-state outputs, ASIL support	Protects independent safety domain and brownout handling	PMIC safety manual/WCCA	RFQ
Power	Application SoC PMIC	1 set	SoC-vendor PMIC/reference power tree	Automotive multiphase PMIC set	Transient response, sequencing, efficiency, safety monitors, low-power state	HPA SoC has complex rail sequencing and load steps	Vendor reference design/PI model	RFQ
Memory	LPDDR5/ LPDDR5X	16-32 GB class	Automotive LPDDR5X	Automotive LPDDR5	Bandwidth, ECC/path support, temp, lifecycle	Cockpit/ virtualization memory capacity and BW	Memory vendor SI/model/qualificat ion	RFQ
Storage	UFS / NVMe	128-512 GB	Automotive UFS 3.x/4.x or NVMe	Automotive eMMC + secondary storage	A/B OTA, logs, endurance, power-loss behavior	Supports lifecycle update/logging	Endurance, temp, power-cut qualification	RFQ
Memory	Safety NVM	1	MCU internal flash + external safety-grade NVM optional	FRAM/MRAM class	Endurance, retention, ECC, power loss	Safety calibration/event metadata	Qualification/ endurance	Design
Network	Central Ethernet switch	1	Marvell Brightlane Q622x class	NXP/other automotive TSN	1/2.5/10G, TSN, MACsec, VLAN,	Future-proofs central gateway	Switch datasheet, thermal,	RFQ

Domain	Item	Qty	Primary/Class	Alternate	Selection Criteria	Justification	Required Evidence	Status
				switch	QoS, diagnostics	beyond 1G baseline	TSN/MACsec	
Network	100/1000BASE-T1 PHY	2-6	Brightlane 88Q2112 class	Other OPEN Alliance automotive PHY	100/1000T1, TC10, 802.1AS, fail-safe, temp	Vehicle Ethernet links	PHY/channel mask/EMC	RFQ
Network	CAN FD transceiver	4-8	TI TCAN1043A/HG-Q1 class	NXP TJA146x class	CAN FD, wake/INH, bus fault, low-power, CAN SIC if needed	Separate powertrain/chassis/body/service networks	ISO11898 compliance, EMC	RFQ
Network	LIN transceiver	0-2	Automotive LIN class	Integrated SBC	Wake, low-power, local console	Physical mode switch/local peripherals if required	EMC/network test	Optional
Security	HSM / secure element	Integrated + optional 1	MCU/SoC integrated HSM	External automotive secure element/TPM	Secure boot, key storage, monotonic counters, crypto throughput	Protects keys and update/command auth	Security target/certification	RFQ
Clock	Automotive oscillator / RTC	2+	Low-ppm automotive oscillator	TCXO class	Jitter, temp drift, startup, safety clock independence	Independent safety and Ethernet clocks	AEC-Q100 / timing analysis	RFQ
Input Protection	Reverse battery / surge front end	1	Automotive ideal-diode/eFuse + TVS	Discrete MOSFET + controller	ISO16750/7637/ OEM transients, low loss, diagnostics	Protects central ECU LV input	Transient DVP/WCCA	Design
Power	Hold-up energy	1	Bulk/polymer capacitor bank	Supercap optional if needed	50ms safety-domain hold-up, ESR, temp, lifetime	Allows safe-state/log on power collapse	Energy calculation/aging	Design
Thermal	Heat spreader/heatsink	1	Conduction plate / liquid cold plate if platform provides	Forced air heat sink	<=75W peak, acoustic, ambient, packaging	Centralized HPA thermal density	CFD/chamber	Design
Display	Display SerDes/DP bridge	2-4	Automotive DP/eDP/FPD-Link/GMSL class	APIX class	Display bandwidth, safety monitoring, EMC, cable length	Raw pixels should not consume 1G vehicle Ethernet	Link/SI/EMC	RFQ
Audio	Audio codec / amp interface	1 set	Automotive audio codec/I2S/TDM	SoC integrated	Warning audio path and infotainment quality	Cockpit integration	Audio/HMI DVP	RFQ
PCB	High-density multilayer PCB	1	Automotive HDI 10-16+ layer class	Two-board mezzanine	LPDDR/PCIe/ Ethernet SI, isolation, thermals	Central compute density	Stack-up/SI/PI/ DFM	Design
Connector	Vehicle I/O connector	2-4	Automotive sealed/unsealed connector per location	Supplier alternate	Current, pin count, CAN/Ethernet SI, retention, service	Vehicle interface	LV214/USCAR/ OEM tests	RFQ
Connector	High-speed Ethernet connector	2-6	H-MTD/MATEnet class	OEM approved alternate	1000/2.5/10GT1 channel, EMC, retention	High-speed backbone	Channel S-parameter	RFQ
Connector	Display high-speed connector	2-4	HFM/Mini-FAKRA class	OEM approved alternate	SerDes/DP channel, EMC, vibration	Cockpit displays	Channel/vibration	RFQ
Diagnostics	DoIP service port/gateway path	1	Ethernet service interface	Vehicle gateway external port	Secure diagnostics, factory/service	Programming/ diagnostics	DoIP/UDS tests	Design
Safety	External supervisor optional	1	Automotive safety supervisor	PMIC integrated only	Diverse watchdog/reset/safe outputs	Additional common-cause mitigation if safety case needs	Safety analysis	Option
EMC	Common-mode chokes/filtering	Per interface	Automotive CAN/Ethernet/filter components	No-CMC where SI permits	CISPR25/ ISO11452 vs SI tradeoff	EMC closure	EMC/SI DVP	Design
Protection	ESD/TVS array	Per external interface	Automotive low-capacitance protection	Integrated PHY protection	ISO10605/OEM ESD, line capacitance	External pin robustness	ESD/SI validation	Design
Mechanical	ECU enclosure	1	Die-cast aluminum/EMI sealed	Stamped/hybrid	Thermal, EMC shielding, crash mounting, serviceability	Central ECU protection	Mechanical/EMC/ thermal	Design
Manufacturing	Secure provisioning fixture	Per line	HSM-backed EOL tool	Offline secure programmer	Keys, VIN, variant, calibration, station audit	Production cybersecurity	EOL validation	Manufacturing
Software	OTA client/bootloader	1	ISO24089/R156-capable update	OEM middleware	Atomic activation, dependency,	Lifecycle update safety	Update DVP	RFQ

Domain	Item	Qty	Primary/Class	Alternate	Selection Criteria	Justification	Required Evidence	Status
			client		recovery			
Software	Diagnostics stack	1	UDS + DoIP stack	Supplier/OEM stack	ISO14229/13400 services, security	Service interoperability	Conformance	RFQ
Software	SOME/IP/ ara::com middleware	1	AUTOSAR Adaptive/SOME-IP class	DDS/OEM service bus	Versioned IPC/network services	Centralized service architecture	Integration/load	RFQ
Software	IDS/firewall	1	Automotive IDS + gateway firewall	OEM security stack	CAN/Ethernet policy, logs, performance	CSMS/TARA controls	Pen/load test	RFQ
Test	HIL rack	1 program	dSPACE SCALEXIO/Micro AutoBox class	NI/Vector HIL class	Plant models, fault injection, timing	SYS4/SYS5 automation	Bench qualification	Program
Test	Network test suite	1 program	Vector CANoe/CANalyzer class	ETAS/other	CAN FD/Ethernet/DoIP/ SOME-IP/SecOC/ E2E	Network verification	Tool qualification as needed	Program
Test	Model-based development	1 program	MATLAB/ Simulink/Stateflow + TargetLink class	Embedded Coder/hand code	Control modeling, code gen, trace, test	Torque/mode/HV algorithms	Tool qualification/validation	Program
ALM	Requirements/ traceability	1 program	IBM DOORS Next / Polarion ALM	Jama/ Codebeamer	Bidirectional ASPICE RTM/change baselines	Process evidence	Tool config/audit	Program
Cyber	PKI/backend integration	1 program	OEM CSMS/SUMS backend	Supplier cloud stack	Certificates, OTA auth, revocation, fleet logs	R155/R156/21434 lifecycle	Pen/audit	Program
Calibration	Calibration toolchain	1 program	CANape/INCA class	OEM calibration tool	A2L/calibration governance, signed release	Mode/torque/ thermal calibration	Calibration audit	Program
Service	Secure service tester	1 program	DoIP/UDS service tester	OEM diagnostic tester	Authenticated routines/update/config	After-sales support	Service validation	Program
Manufacturing	Boundary-scan / ICT	Per line	Production ICT/boundary scan	Functional test only	PCB shorts/opens/programming	Quality and yield	MSA/GRR	Manufacturing
Manufacturing	Thermal/functional run-in	As program requires	Burn-in/run-in station	Sample-based audit	Power, thermal, network, memory stress	Early-life defect screening	Quality study	Manufacturing

8. Development Toolchain and Evidence Environment

The toolchain is treated as a controlled development system rather than a list of preferred brands. Requirements and traceability reside in DOORS Next/Polarion-class ALM; model-based controls use Simulink/Stateflow/TargetLink or equivalent qualified workflows; safety MCU code uses a qualified compiler/MCAL/BSW stack; HPA software uses controlled BSP, hypervisor and Android/Linux/QNX toolchains; verification uses SIL/MIL, HIL, CANoe and vehicle/dyno environments with reproducible configuration.

Framework	Process / Area	Expected Evidence	Package Allocation
Automotive SPICE 4.0	SYS.1/SYS.2/SYS.3; SUP/MAN process interaction	Bidirectional requirements/architecture/test/change evidence	All DOCs + RTM
ISO 26262:2018	Item/HARA/FSC/TSC/system/HW/SW safety	Safety goals, FSR/TSR/HSR/SWSR, analyses, verification	DOC1/DOC2/DOC3
ISO/SAE 21434:2021	Cybersecurity lifecycle/TARA	Assets/threats/goals/controls/ evidence/post-development	DOC2/DOC3
ISO 24089:2023 + Amd1:2024	Software update engineering	Update plan, dependency, vehicle state, activation/recovery evidence	DOC1/DOC3
UNECE R155/R156	Market CSMS/SUMS type-approval obligations where applicable	Cyber/update management evidence and vehicle type trace	DOC1/DOC2
AUTOSAR Classic/Adaptive	MCU BSW, E2E, SecOC, service-oriented architecture	ARXML/BSW/service contracts, safety comm config	DOC3
ISO 11898 CAN/CAN FD Jan-2026 baseline	CAN data-link/physical layer	Network design, timing, conformance	DOC3
ISO 14229 / ISO 13400	UDS / DoIP	Diagnostic service/interface conformance	DOC2/DOC3
ISO 16750-2:2023 / ISO 7637-2:2011	Electrical loads/transients	LV power DVP	DOC3
CISPR 25:2021 / ISO 11452 series	EMC emissions/immunity	Component EMC evidence	DOC3

9. Program Timeline, Resources and Vendor Management

Timing	Major Activities	Gate / Deliverable
M0-M2	Item definition, legacy ECU inventory, stakeholder interviews, RA, ODD/use cases, preliminary HARA/TARA, compute/network/power feasibility	Concept Gate G0
M3-M5	SYS.1/SYS.2, FSC, E/E boundaries, SoC/safety MCU/hypervisor RFQ, initial ICD and traffic/timing analysis	Architecture Gate G1
M6-M8	SYS.3/HWE.1/SWE.1, Sample-A PCB, BSP/MCAL/hypervisor bring-up, SIL/MIL and HIL plant models	Sample A / CDR
M9-M12	Torque/regen/HV/mode algorithms, Android/QNX/Linux integration, CAN/Ethernet gateway, initial cyber controls	Integration Gate G2
M13-M16	Sample-B integrated vehicle prototypes, fault injection, thermal/EMC pre-compliance, network/load stress	DV readiness
M17-M20	Design Validation: HIL, dyno, brake/chassis, HV, electrical, EMC, thermal, cybersecurity, OTA	DV Gate G3
M21-M24	Sample-C production-intent units, PV/EOL station validation, vehicle homologation support, R155/R156 evidence	PV Gate G4
M25-M27	Fleet/road robustness, software/calibration freeze, supplier PPAP/production readiness, service/update release	SOP Readiness G5
M28-M30	SOP ramp, field monitoring, FRACAS, first OTA maintenance release planning	Launch / Post-SOP

9.1 Peak resource plan - 45 FTE reference

Discipline	Peak FTE	Scope
Systems / Architecture	5	Item definition, SYS1/2/3, E/E, ICD, RTM, requirements/change
Functional Safety	4	HARA/FSC/TSC, safety case, FMEA/FTA/FMEDA, HIL fault campaign
VCU Control Software	10	Torque, regen, HV, thermal, driver mode, diagnostics, state management
Safety MCU / BSW	4	AUTOSAR MCAL/BSW, watchdog, communication, safety mechanisms
Cockpit / Infotainment Software	7	Android/QNX/Linux, HMI, service middleware, display integration
Hardware / PCB / SI-PI / EMC	5	Power, MCU/SoC, memory, network, PCB, thermal, EMC
Cybersecurity / OTA	3	TARA, HSM, SecOC, gateway, update, penetration
Validation / HIL / Vehicle	5	SIL/HIL, network, dyno/chassis, EMC/electrical, vehicle DV/PV
Program / Supplier / Quality	2	Planning, RFQ, supplier, PPAP, EOL, release governance

9.2 Vendor evidence and management cadence

Vendor Category	Engagement Flow	Mandatory Evidence	Cadence
Safety MCU / PMIC	RFI -> safety manual review -> benchmark -> A-sample -> FMEDA assumptions -> B-sample freeze	Safety manual/FMEDA/ASIL capability, MCAL, cybersecurity, PCN/EOL, temperature, toolchain	Weekly technical / monthly management
Cockpit SoC	Workload benchmark -> graphics/display -> hypervisor/BSP -> power/thermal -> lifecycle -> nomination	CPU/GPU/memory BW, multi-display, Android/QNX/Linux, safety island, security, supply roadmap	Weekly bring-up / monthly steering
Hypervisor/OS	Certification scope -> supported SoC -> device assignment -> latency -> defect handling -> license	ISO26262 evidence, safety manual, BSP, defect notification, tool qualification	Biweekly
Inverter/BMS/ESC/EPS	ICD freeze -> HIL model -> fault	Signal definitions, limits, timeouts, safety	Weekly cross-functional

Vendor Category	Engagement Flow	Mandatory Evidence	Cadence
	behavior -> timing -> vehicle integration	assumptions, diagnostics, calibration	
Network silicon	Traffic matrix -> switch/PHY selection -> TSN/SecOC/MACsec -> SI/EMC	Datasheets/models, TC10/802.1AS, EMC, long-term supply	Biweekly
Manufacturing/service	EOL secure provisioning -> fixtures -> MSA -> service tester -> OTA release	Station cybersecurity, golden references, service/update process	Monthly / ramp weekly

10. Risk Register

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
RISK-VCU-001	Safety	ASIL C torque target conflicts with HARA ASIL D unintended-torque goal	Under-scoped safety allocation	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-002	Architecture	Single central ECU creates common-cause loss between cockpit and vehicle services	Excess consolidation	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-003	Compute	Chosen HPA SoC not mature enough for SOP timing	Sampling/ roadmap risk	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-004	Thermal	75 W peak budget insufficient for premium cockpit workload	GPU/AI/ display load growth	4	5	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-005	Network	1 Gbps Ethernet becomes bottleneck for future services	Centralization traffic growth	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-006	Software	Android/Linux restart affects shared services	Mixed-criticality integration	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
								evidence; maintain change-controlled residual risk.		
RISK-VCU-007	Hypervisor	Safety hypervisor/device assignment lacks required certification on chosen SoC	Platform support gap	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-008	Safety	Mode transition jerk target violated at large torque delta	50 ms interpretation too aggressive	5	5	25	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-009	Safety	Track mode calibration compromises road safety/homologation	Feature governance	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-010	Chassis	ESC/EPS supplier interfaces do not support requested mode semantics	Supplier mismatch	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-011	Brake	Regen/friction blending ownership unclear	VCU vs ESC boundary	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-012	BMS	HV contactor ownership duplicated	VCU/BMS boundary ambiguity	3	5	15	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-013	Thermal	Hard-coded 65C thresholds incompatible	Calibration oversimplification	4	4	16	High	Allocate owner; close architecture decision early;	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
		with supplier limits						add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.		
RISK-VCU-014	Cyber	Infotainment compromise reaches safety CAN	Gateway isolation weakness	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-015	Cyber	SecOC overhead breaks 10ms CAN timing	Security/ performance co-design	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-016	OTA	Partial multi-domain update creates incompatible versions	Dependency management	4	5	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-017	Diagnostics	DoIP/service routine authorization insufficient	Service attack surface	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-018	Power	Low-voltage brownout resets central ECU while vehicle moving	LV resilience	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-019	EMC	High-speed SoC/DDR/SerDes noise fails CISPR25	Central compute EMC density	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
RISK-VCU-020	PI	HPA load-step causes DDR/SoC rail droop	Power integrity	5	5	25	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-021	SI	Display/Ethernet/DDR SI fails on compact PCB	High-speed layout	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-022	Manufacturing	Secure provisioning station compromised	Production cyber	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-023	Quality	Wrong variant/calibration at EOL	Configuration management	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-024	Lifecycle	Critical SoC/hypervisor supplier EOL before vehicle service life	Obsolescence	3	5	15	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-025	Toolchain	Model/code-generation/tool qualification evidence incomplete	ISO26262 tool confidence	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-026	ASPICE	Requirements duplicated across SYS2/SYS3 without ownership	Traceability quality	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
								change-controlled residual risk.		
RISK-VCU-027	HIL	Plant models do not cover coupled brake/thermal/HV states	Verification coverage	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-028	Timing	Worst-case CAN arbitration exceeds torque deadline	Network schedulability	4	5	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-029	Storage	Flight recorder/OTA writes exhaust storage endurance	Lifetime	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-030	Boot	Cockpit slow boot hides safety state	User/safety perception	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-031	HMI	Driver sees requested mode rather than authoritative accepted mode	State mismatch	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-032	Security	Key/certificate expiry disrupts service/update	Lifecycle PKI	5	5	25	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-033	Performance	220 kW peak requested when BMS SOP lower	Limit arbitration	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/D	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
								VP and supplier evidence; maintain change-controlled residual risk.		
RISK-VCU-034	Vehicle Dynamics	Torque rate limit conflicts with emergency cut	Comfort vs safety	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-035	Service	Actuator routine accidentally allowed in Drive	Diagnostic state gating	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-036	Supplier	Inverter/BMS limit signals have inconsistent units/age	ICD quality	3	5	15	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-037	Change	Track feature late change causes safety/cyber regression	Feature creep	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-038	Cost	Dual compute architecture exceeds BOM target	Cost pressure	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-039	Power	Dual SoC/MCU PMIC architecture exceeds 75 W	Power budget	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-040	Cooling	Cabin mounting	Packaging	4	5	20	Critical	Allocate owner; close	Domain Lead	Open

Risk ID	Category	Risk	Cause	P	I	Score	Level	Mitigation	Owner	Status
		cannot reject peak heat silently						architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.		
RISK-VCU-041	Privacy	Fleet logs contain personal driver data	Data governance	5	4	20	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-042	Regulation	UNECE R155/R156 market scope addressed too late	Homologation	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-043	Testing	EMC test modes do not exercise worst GPU/network activity	False pass	4	4	16	High	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-044	Safety	Independent safety MCU shares reset/clock/power root with HPA	Common-cause	5	5	25	Critical	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open
RISK-VCU-045	Calibration	User custom mode parameters exceed safe bounds	Customization control	3	4	12	Medium	Allocate owner; close architecture decision early; add targeted analysis/HIL/DVP and supplier evidence; maintain change-controlled residual risk.	Domain Lead	Open

11. Requirement Authority (RA) Baseline

The RA layer freezes stakeholder-visible capability and non-negotiable architectural intent before decomposition into SYS.1/SYS.2. Every RA item is expected to remain testable or traceable to an analysis/approval artifact. Any later platform-specific relaxation is handled through change control rather than silent SYS.3 divergence.

RA ID	Area	Requirement	Priority	Verification
RA-VCU-001	Safety & Motion	The central ECU shall prevent a QM cockpit/infotainment fault or compromise from creating an unvalidated positive traction torque request.	Must	Safety analysis + fault injection
RA-VCU-002	Driver Modes	Driver-mode changes shall be validated against vehicle state, thermal/SOP limits, market configuration and chassis permissions before they affect torque/regen/chassis requests.	Must	HIL + vehicle transition test
RA-VCU-003	HV Coordination	VCU HV orchestration shall coordinate BMS, inverter and charger states while pack contactor actuation and pack protection remain owned by the BMS unless a vehicle item definition explicitly reallocates ownership.	Must	Interface review + HIL
RA-VCU-004	Thermal & Energy	No driver mode shall increase torque or regen above the minimum of BMS SOP, inverter/motor capability, thermal limits, traction limits and legal/market caps.	Must	HIL boundary sweep
RA-VCU-005	Chassis & Regen	Regenerative braking coordination shall remain subordinate to ESC/ABS wheel-slip and friction-brake safety authority.	Must	Brake HIL + vehicle
RA-VCU-006	Central Compute	The architecture shall use a physically independent safety MCU or equivalently independent safety island for final motion-authority supervision even when IVI and vehicle services share a central SoC/hypervisor.	Must	Safety concept review
RA-VCU-007	Networking	The VCU220 program shall provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-008	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-009	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-010	Cybersecurity	The VCU220 program shall prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-011	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-012	Lifecycle/Quality	The VCU220 program shall	Must	Architecture/requirements

RA ID	Area	Requirement	Priority	Verification
		support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.		review + allocated verification
RA-VCU-013	Safety & Motion	The VCU220 program shall prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-014	Driver Modes	The VCU220 program shall support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-015	HV Coordination	The VCU220 program shall coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-016	Thermal & Energy	The VCU220 program shall enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-017	Chassis & Regen	The VCU220 program shall coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-018	Central Compute	The VCU220 program shall consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-019	Networking	The VCU220 program shall provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-020	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-021	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 2 shall be verified at	Must	Architecture/requirements review + allocated verification

RA ID	Area	Requirement	Priority	Verification
		the appropriate model, bench, HIL, vehicle or production gate.		
RA-VCU-022	Cybersecurity	The VCU220 program shall prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-023	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-024	Lifecycle/Quality	The VCU220 program shall support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-025	Safety & Motion	The VCU220 program shall prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-026	Driver Modes	The VCU220 program shall support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-027	HV Coordination	The VCU220 program shall coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-028	Thermal & Energy	The VCU220 program shall enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-029	Chassis & Regen	The VCU220 program shall coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-030	Central Compute	The VCU220 program shall consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-031	Networking	The VCU220 program shall provide deterministic CAN FD	Must	Architecture/requirements review + allocated verification

RA ID	Area	Requirement	Priority	Verification
		motion networks and scalable Ethernet/TSN service backbone; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.		
RA-VCU-032	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-033	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-034	Cybersecurity	The VCU220 program shall prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-035	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-036	Lifecycle/Quality	The VCU220 program shall support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-037	Safety & Motion	The VCU220 program shall prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-038	Driver Modes	The VCU220 program shall support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-039	HV Coordination	The VCU220 program shall coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-040	Thermal & Energy	The VCU220 program shall enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 4 shall be verified at the appropriate model, bench,	Must	Architecture/requirements review + allocated verification

RA ID	Area	Requirement	Priority	Verification
		HIL, vehicle or production gate.		
RA-VCU-041	Chassis & Regen	The VCU220 program shall coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-042	Central Compute	The VCU220 program shall consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-043	Networking	The VCU220 program shall provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-044	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-045	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-046	Cybersecurity	The VCU220 program shall prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-047	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-048	Lifecycle/Quality	The VCU220 program shall support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-049	Safety & Motion	The VCU220 program shall prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-050	Driver Modes	The VCU220 program shall support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement	Must	Architecture/requirements review + allocated verification

RA ID	Area	Requirement	Priority	Verification
		instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.		
RA-VCU-051	HV Coordination	The VCU220 program shall coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-052	Thermal & Energy	The VCU220 program shall enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-053	Chassis & Regen	The VCU220 program shall coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-054	Central Compute	The VCU220 program shall consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-055	Networking	The VCU220 program shall provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-056	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-057	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-058	Cybersecurity	The VCU220 program shall prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-059	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Must	Architecture/requirements review + allocated verification
RA-VCU-060	Lifecycle/Quality	The VCU220 program shall support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and	Must	Architecture/requirements review + allocated verification

RA ID	Area	Requirement	Priority	Verification
		production quality; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.		

12. Preliminary Functional Safety Concept

ASIL boundary: The program target states ASIL C for torque arbitration, while the preliminary unintended-torque safety goal is conservatively treated as ASIL D. This is not a contradiction to hide: the final vehicle HARA/FSC/TSC shall determine the item ASIL and decomposition. The architecture therefore preserves an ASIL-D-capable safety-control path.

SG	Safety Goal	Preliminary ASIL	Concept
SG-01	Prevent unintended or excessive positive propulsion torque caused by VCU malfunction or invalid mode/calibration.	Preliminary ASIL D	Independent torque supervision, plausibility, bounded communication and inverter local safety.
SG-02	Prevent hazardous loss or reversal of commanded propulsion torque that could destabilize the vehicle.	Preliminary ASIL C	Degraded drive, driver warning and controlled stop strategy.
SG-03	Prevent hazardous regenerative braking or deceleration request inconsistent with driver/brake/ESC state.	Preliminary ASIL C	Brake authority, regen clamp and independent plausibility.
SG-04	Prevent an unsafe HV enable/contact state request or propulsion enable before valid precharge/HV readiness.	Preliminary ASIL C	BMS-owned contactors, VCU coordination, dual permissives.
SG-05	Prevent driver-mode corruption from increasing torque/regen/chassis authority beyond released limits.	Preliminary ASIL B/C	Safety-domain validation and signed calibration.
SG-06	Prevent QM infotainment/Android faults from interfering with safety motion-control resources.	Preliminary ASIL C	Independent safety MCU, MPU/IOMMU, watchdog and gateway filtering.
SG-07	Prevent loss of critical driver warning/telltale information during safety-relevant VCU faults.	Preliminary ASIL B	Independent monitored telltale path and degraded indication.
SG-08	Prevent thermal/energy-limit violation that can lead to loss of propulsion, battery or inverter damage.	Preliminary ASIL B/C	Minimum capability arbitration and monotonic derating.
SG-09	Prevent corrupted/stale network data from increasing motion/HV capability.	Preliminary ASIL C	Age/E2E/SecOC/timeout non-permissive behavior.
SG-10	Prevent unauthorized software/calibration/update from modifying safety-relevant vehicle-control behavior.	Security-to-Safety	Secure boot/update, HSM, signed calibration and safety re-validation.
SG-11	Prevent unsafe central-ECU behavior due to LV power, thermal, clock or memory failure.	Preliminary ASIL C	Independent PMIC/watchdog, ECC, thermal and brownout safe state.
SG-12	Prevent manufacturing/service configuration mismatch from enabling incompatible vehicle functions.	Preliminary ASIL B	Secure EOL binding, topology/variant checks and commissioning tests.

12.1 Functional Safety Requirements

FSR	SG	Functional Safety Requirement	FTTI / Timing	Allocation
FSR-VCU-001	SG-01	Authoritative propulsion torque shall be computed/validated in the safety domain and independently bounded before transmission.	<=10 ms internal reaction	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-002	SG-01	Brake application, invalid gear, crash or severe plausibility fault shall revoke positive torque authority.	<=10 ms	Safety MCU / central compute / network / external ECU as allocated

FSR	SG	Functional Safety Requirement	FTTI / Timing	Allocation
FSR-VCU-003	SG-02	Loss of primary control shall transition to deterministic degraded torque/controlled stop rather than arbitrary output.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-004	SG-03	Regen command shall be subordinate to ESC/ABS/brake authority and charge-SOP constraints.	10 ms command cycle	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-005	SG-04	HV propulsion enable shall require valid BMS HV-ready/precharge and inverter readiness.	Before enable	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-006	SG-05	Driver-mode request shall be treated as untrusted until safety-domain validation succeeds.	<=50 ms acknowledge	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-007	SG-06	QM partitions shall have no direct write access to safety actuation peripherals or safety CAN controllers.	Continuous	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-008	SG-07	Critical safety telltales shall be independently monitorable from Android rich-HMI execution.	<=100 ms indication	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-009	SG-08	Torque availability shall be the minimum of BMS SOP, inverter/motor limits, thermal limits and safety caps.	Each 10 ms cycle	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-010	SG-09	Stale/invalid safety messages shall be non-permissive and shall not increase vehicle authority.	Message-specific timeout	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-011	SG-10	Boot/update/calibration shall authenticate code/data and reject unauthorized rollback/configuration.	Before activation	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-012	SG-11	Independent PMIC/watchdog shall force safe output on MCU rail/clock/task faults.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-013	SG-12	Variant/topology/configuration mismatch shall block incompatible safety feature activation.	Before READY	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-014	SG-01	Authoritative propulsion torque shall be computed/validated in the safety domain and independently bounded before transmission.	<=10 ms internal reaction	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-015	SG-01	Brake application, invalid gear, crash or severe plausibility fault shall revoke positive torque authority.	<=10 ms	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-016	SG-02	Loss of primary control shall transition to deterministic degraded torque/controlled stop rather than arbitrary output.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-017	SG-03	Regen command shall be subordinate to ESC/ABS/brake authority and charge-SOP constraints.	10 ms command cycle	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-018	SG-04	HV propulsion enable shall require valid BMS HV-ready/precharge and inverter readiness.	Before enable	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-019	SG-05	Driver-mode request shall be treated as untrusted until safety-domain validation succeeds.	<=50 ms acknowledge	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-020	SG-06	QM partitions shall have no direct write access to safety actuation peripherals or safety CAN controllers.	Continuous	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-021	SG-07	Critical safety telltales shall be independently monitorable from Android rich-HMI execution.	<=100 ms indication	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-022	SG-08	Torque availability shall be the	Each 10 ms cycle	Safety MCU / central compute /

FSR	SG	Functional Safety Requirement	FTTI / Timing	Allocation
		minimum of BMS SOP, inverter/motor limits, thermal limits and safety caps.		network / external ECU as allocated
FSR-VCU-023	SG-09	Stale/invalid safety messages shall be non-permissive and shall not increase vehicle authority.	Message-specific timeout	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-024	SG-10	Boot/update/calibration shall authenticate code/data and reject unauthorized rollback/configuration.	Before activation	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-025	SG-11	Independent PMIC/watchdog shall force safe output on MCU rail/clock/task faults.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-026	SG-12	Variant/topology/configuration mismatch shall block incompatible safety feature activation.	Before READY	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-027	SG-01	Authoritative propulsion torque shall be computed/validated in the safety domain and independently bounded before transmission.	<=10 ms internal reaction	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-028	SG-01	Brake application, invalid gear, crash or severe plausibility fault shall revoke positive torque authority.	<=10 ms	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-029	SG-02	Loss of primary control shall transition to deterministic degraded torque/controlled stop rather than arbitrary output.	FTTI-derived	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-030	SG-03	Regen command shall be subordinate to ESC/ABS/brake authority and charge-SOP constraints.	10 ms command cycle	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-031	SG-04	HV propulsion enable shall require valid BMS HV-ready/precharge and inverter readiness.	Before enable	Safety MCU / central compute / network / external ECU as allocated
FSR-VCU-032	SG-05	Driver-mode request shall be treated as untrusted until safety-domain validation succeeds.	<=50 ms acknowledge	Safety MCU / central compute / network / external ECU as allocated

13. Cybersecurity and Software-Update Concept

The cockpit/Android domain is treated as exposed attack surface. Safety authority is protected by hardware isolation, HSM-backed trust anchors, gateway filtering, authenticated diagnostics/update, network segmentation and independent safety validation. AUTOSAR E2E is used for communication-safety integrity; SecOC is allocated independently where freshness/authenticity are required by TARA.

14. Reference and Vendor Evidence Register

Ref ID	Category	Document / Product	Source	Link	Use
REF-VCU-001	Process	Automotive SPICE 4.0 PRM/PAM	VDA QMC	Official source	SYS.1/SYS.2/SYS.3 process and assessment baseline
REF-VCU-002	Functional Safety	ISO 26262:2018 Road vehicles - Functional safety	ISO	Official source	Item/HARA/FSC/TSC/ system/HW/SW safety lifecycle
REF-VCU-003	Cybersecurity	ISO/SAE 21434:2021 Road vehicles - Cybersecurity engineering	ISO	Official source	TARA and cybersecurity lifecycle
REF-VCU-004	Software Update	ISO 24089:2023 + Amd 1:2024	ISO	Official source	Software update engineering
REF-VCU-005	Regulation	UN Regulation No.155 - Cyber security / CSMS	UNECE	Official source	Market cybersecurity type-approval comparator
REF-VCU-006	Regulation	UN Regulation No.156 - Software update / SUMS	UNECE	Official source	Market software-update type-approval comparator

Ref ID	Category	Document / Product	Source	Link	Use
REF-VCU-007	Diagnostics	ISO 14229-1:2020 UDS (Jan-2026 baseline)	ISO	Official source	UDS diagnostic service architecture
REF-VCU-008	Diagnostics	ISO 13400-2:2025 DoIP (Jan-2026 baseline)	ISO	Official source	IP diagnostic transport
REF-VCU-009	CAN	ISO 11898-1:2024 / ISO 11898-2:2024 Jan-2026 baseline	ISO	Official source	CAN/CAN FD data-link and high-speed physical layer
REF-VCU-010	Electrical	ISO 16750-2:2023 Electrical loads	ISO	Official source	LV electrical environmental loads
REF-VCU-011	Electrical	ISO 7637-2:2011 Conducted transients	ISO	Official source	12/24 V transient bench test baseline
REF-VCU-012	EMC	CISPR 25:2021	IEC	Official source	Component conducted/radiated emissions
REF-VCU-013	EMC	ISO 11452-2:2019 / ISO 11452-4:2020	ISO	Official source	Component radiated/BCI immunity
REF-VCU-014	AUTOSAR	AUTOSAR E2E Protocol Specification	AUTOSAR	Official source	Communication safety integrity profiles
REF-VCU-015	AUTOSAR	AUTOSAR Secure Onboard Communication (SecOC)	AUTOSAR	Official source	PDU authenticity and freshness
REF-VCU-016	Safety MCU	AURIX TC4x family	Infineon	Official source	Safety MCU / eMobility / vehicle motion control reference
REF-VCU-017	Safety MCU PMIC	OPTIREG TLF35585	Infineon	Official source	Safety PMIC/watchdog/rail/safe-state reference
REF-VCU-018	Central real-time compute	S32N55 / S32N family	NXP	Official source	Central real-time vehicle control / hardware isolation comparator
REF-VCU-019	HPA SoC	R-Car X5H / Gen5 central compute	Renesas	Official source	Multi-domain HPA / IVI / virtualization comparator
REF-VCU-020	Cockpit SoC	Snapdragon Cockpit Elite	Qualcomm	Official source	Premium virtualized cockpit/central compute comparator
REF-VCU-021	Hypervisor	QNX Hypervisor for Safety	QNX	Official source	Safety-qualified virtualization comparator
REF-VCU-022	Ethernet Switch	Brightlane Q622x central automotive Ethernet switch	Marvell	Official source	60/90-Gbps TSN/high-speed central switching comparator
REF-VCU-023	Ethernet PHY	Brightlane 88Q2110/88Q2112	Infineon/Marvell technology line	Official source	100/1000BASE-T1 PHY / TC10 / 802.1AS comparator
REF-VCU-024	CAN PHY	TCAN1043A/HG-Q1 CAN FD transceiver	Texas Instruments	Official source	CAN FD / wake / bus fault comparator
REF-VCU-025	Cockpit	Snapdragon Cockpit Platform	Qualcomm	Official source	Cockpit virtualized processing / UX ecosystem
REF-VCU-026	Central Compute	R-Car X5H architecture overview	Renesas	Official source	Centralized multi-domain SoC capabilities
REF-VCU-027	Software	AUTOSAR Standards	AUTOSAR	Official source	Classic/Adaptive architecture baseline

Appendix A - Block Ownership Map

Block	Controlled Architecture Scope
B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control
B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway
B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization
B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs
B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager
B06	Regenerative Braking, Deceleration Coordination & Brake-Energy

Block	Controlled Architecture Scope
	Manager
B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control
B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager
B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks
B10	Energy, Thermal, Performance Limit & Derating Supervisor
B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management
B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness
B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary
B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging
B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management
B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services
B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management
B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design
B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning
B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS

Appendix B - Traceability and Release Model

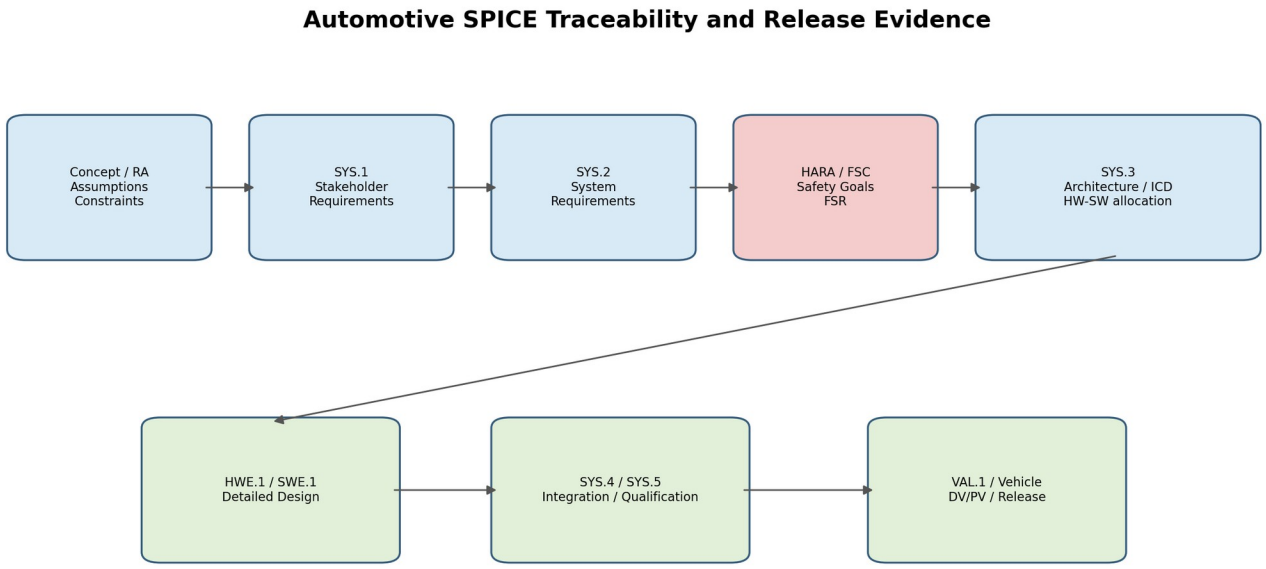


Figure 6 - Automotive SPICE traceability and evidence chain

DOC1 freezes concept, RA, assumptions, initial safety/cyber architecture and the program execution model. DOC2 decomposes SYS.1/SYS.2 and safety/risk/interface analyses. DOC3 owns SYS.3 architecture, HW/SW allocation and verification handoff. The master workbook is the controlled tabular RTM/DVP source and is part of the release package.

Appendix C - Requirement Authority Technical Interpretation

This appendix converts each RA statement into the engineering interpretation expected at architecture review and the evidence that should close the requirement. It prevents a future program from reusing the RA identifiers while silently changing their safety or verification intent.

RA ID	Area	Frozen Intent	Technical Interpretation	Closure Evidence
RA-VCU-001	Safety & Motion	The central ECU shall prevent a QM cockpit/infotainment fault or compromise from creating an unvalidated positive traction torque request.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.
RA-VCU-002	Driver Modes	Driver-mode changes shall be validated against vehicle state, thermal/SOP limits, market configuration and chassis permissions before they affect torque/regen/chassis requests.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.
RA-VCU-003	HV Coordination	VCU HV orchestration shall coordinate BMS, inverter and charger states while pack contactor actuation and pack protection remain owned by the BMS unless a vehicle item definition explicitly reallocates ownership.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.
RA-VCU-004	Thermal & Energy	No driver mode shall increase torque or regen above the minimum of BMS SOP, inverter/motor capability, thermal limits, traction limits and legal/market caps.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.
RA-VCU-005	Chassis & Regen	Regenerative braking coordination shall remain subordinate to ESC/ABS wheel-slip and friction-brake safety authority.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.
RA-VCU-006	Central Compute	The architecture shall use a physically independent safety MCU or equivalently independent safety island for final motion-authority supervision even when IVI and vehicle services share a central SoC/hypervisor.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.
RA-VCU-007	Networking	The VCU220 program shall provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.
RA-VCU-008	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.
RA-VCU-009	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Diagnostic and plant/service functions require vehicle-state interlocks, role authorization and complete audit/freeze-frame data. Service modes cannot bypass torque/HV safety.	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.
RA-VCU-010	Cybersecurity	The VCU220 program shall prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Protect roots of trust, boot/update/calibration, diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.
RA-VCU-011	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Close power, transient, thermal, SI/PI and EMC margins on the complete central ECU. External display/audio power is separately budgeted unless intentionally included.	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.
RA-VCU-012	Lifecycle/Quality	The VCU220 program shall support ASPICE traceability,	Maintain serialized configuration, manufacturing evidence,	EOL station MSA, secure provisioning, PV audits, supplier

RA ID	Area	Frozen Intent	Technical Interpretation	Closure Evidence
		supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 1 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	PCN/EOL reviews, field FRACAS and post-SOP update evidence.
RA-VCU-013	Safety & Motion	The VCU220 program shall prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.
RA-VCU-014	Driver Modes	The VCU220 program shall support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.
RA-VCU-015	HV Coordination	The VCU220 program shall coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.
RA-VCU-016	Thermal & Energy	The VCU220 program shall enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.
RA-VCU-017	Chassis & Regen	The VCU220 program shall coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.
RA-VCU-018	Central Compute	The VCU220 program shall consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.
RA-VCU-019	Networking	The VCU220 program shall provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.
RA-VCU-020	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.
RA-VCU-021	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Diagnostic and plant/service functions require vehicle-state interlocks, role authorization and complete audit/freeze-frame data. Service modes cannot bypass torque/HV safety.	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.
RA-VCU-022	Cybersecurity	The VCU220 program shall prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Protect roots of trust, boot/update/calibration, diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.
RA-VCU-023	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement	Close power, transient, thermal, SI/PI and EMC margins on the complete central ECU. External display/audio power is separately	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.

RA ID	Area	Frozen Intent	Technical Interpretation	Closure Evidence
		instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	budgeted unless intentionally included.	
RA-VCU-024	Lifecycle/Quality	The VCU220 program shall support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 2 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Maintain serialized configuration, manufacturing evidence, PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.
RA-VCU-025	Safety & Motion	The VCU220 program shall prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.
RA-VCU-026	Driver Modes	The VCU220 program shall support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.
RA-VCU-027	HV Coordination	The VCU220 program shall coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.
RA-VCU-028	Thermal & Energy	The VCU220 program shall enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.
RA-VCU-029	Chassis & Regen	The VCU220 program shall coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.
RA-VCU-030	Central Compute	The VCU220 program shall consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.
RA-VCU-031	Networking	The VCU220 program shall provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.
RA-VCU-032	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.
RA-VCU-033	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Diagnostic and plant/service functions require vehicle-state interlocks, role authorization and complete audit/freeze-frame data. Service modes cannot bypass torque/HV safety.	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.
RA-VCU-034	Cybersecurity	The VCU220 program shall prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 3 shall be verified at the appropriate model, bench, HIL,	Protect roots of trust, boot/update/calibration, diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.

RA ID	Area	Frozen Intent	Technical Interpretation	Closure Evidence
		vehicle or production gate.		
RA-VCU-035	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Close power, transient, thermal, SI/PI and EMC margins on the complete central ECU. External display/audio power is separately budgeted unless intentionally included.	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.
RA-VCU-036	Lifecycle/Quality	The VCU220 program shall support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 3 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Maintain serialized configuration, manufacturing evidence, PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.
RA-VCU-037	Safety & Motion	The VCU220 program shall prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.
RA-VCU-038	Driver Modes	The VCU220 program shall support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.
RA-VCU-039	HV Coordination	The VCU220 program shall coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.
RA-VCU-040	Thermal & Energy	The VCU220 program shall enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.
RA-VCU-041	Chassis & Regen	The VCU220 program shall coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.
RA-VCU-042	Central Compute	The VCU220 program shall consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.
RA-VCU-043	Networking	The VCU220 program shall provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.
RA-VCU-044	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.
RA-VCU-045	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Diagnostic and plant/service functions require vehicle-state interlocks, role authorization and complete audit/freeze-frame data. Service modes cannot bypass torque/HV safety.	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.
RA-VCU-046	Cybersecurity	The VCU220 program shall prevent unauthorized	Protect roots of trust, boot/update/calibration,	TARA-based penetration, secure boot/update rollback,

RA ID	Area	Frozen Intent	Technical Interpretation	Closure Evidence
		command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	key/certificate, replay/spoofing and compromised-cockpit tests.
RA-VCU-047	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Close power, transient, thermal, SI/PI and EMC margins on the complete central ECU. External display/audio power is separately budgeted unless intentionally included.	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.
RA-VCU-048	Lifecycle/Quality	The VCU220 program shall support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 4 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Maintain serialized configuration, manufacturing evidence, PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.
RA-VCU-049	Safety & Motion	The VCU220 program shall prevent unintended torque, retain deterministic safe/degraded behavior and independent actuation safety; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Treat the requirement as an authority/containment rule. The safety MCU and local actuator ECU shall remain capable of rejecting an unsafe command even if the HPA or HMI behaves incorrectly.	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.
RA-VCU-050	Driver Modes	The VCU220 program shall support controlled Eco/Normal/Sport/Track/Valet profiles with jerk-bounded transition and safety-clamped calibration; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Convert the user-visible mode request into a safety-domain accepted profile. Separate request acknowledgement from physical torque/regen/chassis blending and enforce jerk/SOP/thermal/local-ECU limits.	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.
RA-VCU-051	HV Coordination	The VCU220 program shall coordinate vehicle HV enable/precharge/drive/charge states without replacing BMS pack protection; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Keep pack contactor/precharge actuation in the BMS reference allocation while the VCU orchestrates vehicle READY and inverter enable. Every transition uses explicit prerequisites and feedback.	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.
RA-VCU-052	Thermal & Energy	The VCU220 program shall enforce BMS/inverter/motor/charger power limits and prioritize cooling without violating component limits; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Use normalized capability/SOP limits from the BMS, inverter, motor and thermal controller rather than a hard-coded universal temperature threshold. Capability may only decrease on invalid/stale input.	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.
RA-VCU-053	Chassis & Regen	The VCU220 program shall coordinate regenerative braking and chassis requests while preserving ESC/ABS/EPS authority; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	The VCU calculates desired regen/chassis profile but the ESC/brake/EPS controllers retain local wheel-slip, hydraulic and steering safety authority.	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.
RA-VCU-054	Central Compute	The VCU220 program shall consolidate cockpit and vehicle services with independent safety MCU and hardware-enforced isolation; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Prove freedom from interference across safety MCU, hypervisor, HPA guests, memory, power and network resources. Authoritative motion control must survive HPA reset or overload.	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.
RA-VCU-055	Networking	The VCU220 program shall provide deterministic CAN FD motion networks and scalable Ethernet/TSN service backbone; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Specify timing, validity, age, routing and security separately for safety CAN and IP/Ethernet services. Ethernet must not be a single point of failure for motion safety.	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.
RA-VCU-056	Cockpit/HMI	The VCU220 program shall provide responsive driver interaction and safety-telltale independence from rich infotainment; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Treat rich HMI as a presentation/request layer. Safety-relevant telltales and mode status require monitored source integrity and a defined fallback if Android/display rendering fails.	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.
RA-VCU-057	Diagnostics/Service	The VCU220 program shall support UDS/DoIP, event recording, calibration, EOL and service workflows; requirement	Diagnostic and plant/service functions require vehicle-state interlocks, role authorization and complete audit/freeze-frame	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.

RA ID	Area	Frozen Intent	Technical Interpretation	Closure Evidence
		instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	data. Service modes cannot bypass torque/HV safety.	
RA-VCU-058	Cybersecurity	The VCU220 program shall prevent unauthorized command/update/configuration paths and support lifecycle TARA/monitoring; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Protect roots of trust, boot/update/calibration, diagnostics and network trust boundaries. E2E integrity and SecOC authenticity/freshness are treated as distinct mechanisms.	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.
RA-VCU-059	Electrical/EMC	The VCU220 program shall meet LV electrical, EMC, thermal, mechanical and environmental design targets; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Close power, transient, thermal, SI/PI and EMC margins on the complete central ECU. External display/audio power is separately budgeted unless intentionally included.	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.
RA-VCU-060	Lifecycle/Quality	The VCU220 program shall support ASPICE traceability, supplier evidence, OTA, obsolescence, FRACAS and production quality; requirement instance 5 shall be verified at the appropriate model, bench, HIL, vehicle or production gate.	Maintain serialized configuration, manufacturing evidence, PCN/EOL control, FRACAS, cybersecurity support and service/update compatibility through program life.	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.

Appendix D - Component/BOM Technical Evidence Plan

The architecture BOM is not a purchasing list. Each line below defines why the component class exists, what must be demonstrated before nomination, and which evidence is expected from the supplier or internal design team. Exact manufacturer part numbers are frozen only after A-sample benchmark and safety/cyber/lifecycle review.

Domain	Item	Primary/Class	Alternate	Nomination Evidence & Design Rationale	Status
Compute	Safety MCU	Infineon AURIX TC4x class	Renesas RH850/U2B / NXP S32N class	Nomination shall close: Lockstep, ECC, HSM, CAN/Ethernet, safety manual, AUTOSAR ecosystem, lifecycle. Required evidence: Safety manual, FMEDA, MCU benchmark, PCN/EOL. The selected solution must preserve the architecture justification: Independent deterministic authority for torque/HV supervision	RFQ
Compute	Application/Cockpit SoC	Renesas R-Car X5H class	Qualcomm Snapdragon Cockpit Elite class	Nomination shall close: CPU/GPU, virtualization, multi-display, memory BW, power, safety island, Android/Linux/QNX support, lifecycle. Required evidence: Target workload benchmark, thermal, safety/security collateral. The selected solution must preserve the architecture justification: Supports premium cockpit + centralized services while torque remains external on safety MCU	RFQ
Compute	Safety-centric central processor alternate	NXP S32N55 class	AURIX TC4x + gateway FPGA/ASIC	Nomination shall close: Deterministic real-time consolidation, hardware isolation, Ethernet switch/security. Required evidence: NXP safety/performance evidence. The selected solution must preserve the architecture justification: Strong central vehicle control option if premium cockpit remains separate/paired	Trade study
Software	Safety hypervisor	QNX Hypervisor for Safety class	Green Hills INTEGRITY Multivisor class / supplier safety hypervisor	Nomination shall close: ISO26262 evidence, guest isolation, Android/Linux support, device assignment, boot time. Required evidence: Safety certificate/manual, latency, supported SoC. The selected solution must preserve the architecture	RFQ

Domain	Item	Primary/Class	Alternate	Nomination Evidence & Design Rationale	Status
				justification: Reduces mixed-criticality integration risk	
Software	Safety OS/AUTOSAR Classic	AUTOSAR Classic OS/BSW + safety vendor stack	SafeRTOS / proprietary safety RTOS as project architecture permits	Nomination shall close: Timing, MCAL, safety qualification, diagnostics, SecOC/E2E. Required evidence: Safety qualification/toolchain evidence. The selected solution must preserve the architecture justification: Deterministic safety MCU platform	RFQ
Software	Cockpit OS	Android Automotive + Linux/QNX services	QNX cockpit / AGL	Nomination shall close: OEM UX ecosystem, app model, boot time, security, lifecycle. Required evidence: BSP, update, security support. The selected solution must preserve the architecture justification: Rich HMI kept behind safety boundary	RFQ
Power	Safety PMIC	Infineon TLF35585 class	NXP automotive safety PMIC class	Nomination shall close: 3-40V input, watchdog, rail monitoring, safe-state outputs, ASIL support. Required evidence: PMIC safety manual/WCCA. The selected solution must preserve the architecture justification: Protects independent safety domain and brownout handling	RFQ
Power	Application SoC PMIC	SoC-vendor PMIC/reference power tree	Automotive multiphase PMIC set	Nomination shall close: Transient response, sequencing, efficiency, safety monitors, low-power state. Required evidence: Vendor reference design/PI model. The selected solution must preserve the architecture justification: HPA SoC has complex rail sequencing and load steps	RFQ
Memory	LPDDR5/LPDDR5X	Automotive LPDDR5X	Automotive LPDDR5	Nomination shall close: Bandwidth, ECC/path support, temp, lifecycle. Required evidence: Memory vendor SI/model/qualification. The selected solution must preserve the architecture justification: Cockpit/virtualization memory capacity and BW	RFQ
Storage	UFS / NVMe	Automotive UFS 3.x/4.x or NVMe	Automotive eMMC + secondary storage	Nomination shall close: A/B OTA, logs, endurance, power-loss behavior. Required evidence: Endurance, temp, power-cut qualification. The selected solution must preserve the architecture justification: Supports lifecycle update/logging	RFQ
Memory	Safety NVM	MCU internal flash + external safety-grade NVM optional	FRAM/MRAM class	Nomination shall close: Endurance, retention, ECC, power loss. Required evidence: Qualification/endurance. The selected solution must preserve the architecture justification: Safety calibration/event metadata	Design
Network	Central Ethernet switch	Marvell Brightlane Q622x class	NXP/other automotive TSN switch	Nomination shall close: 1/2.5/10G, TSN, MACsec, VLAN, QoS, diagnostics. Required evidence: Switch datasheet, thermal, TSN/MACsec. The selected solution must preserve the architecture justification: Future-proofs central gateway beyond 1G baseline	RFQ
Network	100/1000BASE-T1 PHY	Brightlane 88Q2112 class	Other OPEN Alliance automotive PHY	Nomination shall close: 100/1000T1, TC10,	RFQ

Domain	Item	Primary/Class	Alternate	Nomination Evidence & Design Rationale	Status
				802.1AS, fail-safe, temp. Required evidence: PHY/channel mask/EMC. The selected solution must preserve the architecture justification: Vehicle Ethernet links	
Network	CAN FD transceiver	TI TCAN1043A/HG-Q1 class	NXP TJA146x class	Nomination shall close: CAN FD, wake/INH, bus fault, low-power, CAN SIC if needed. Required evidence: ISO11898 compliance, EMC. The selected solution must preserve the architecture justification: Separate powertrain/chassis/body/service networks	RFQ
Network	LIN transceiver	Automotive LIN class	Integrated SBC	Nomination shall close: Wake, low-power, local console. Required evidence: EMC/network test. The selected solution must preserve the architecture justification: Physical mode switch/local peripherals if required	Optional
Security	HSM / secure element	MCU/SoC integrated HSM	External automotive secure element/TPM	Nomination shall close: Secure boot, key storage, monotonic counters, crypto throughput. Required evidence: Security target/certification. The selected solution must preserve the architecture justification: Protects keys and update/command auth	RFQ
Clock	Automotive oscillator / RTC	Low-ppm automotive oscillator	TCXO class	Nomination shall close: Jitter, temp drift, startup, safety clock independence. Required evidence: AEC-Q100 / timing analysis. The selected solution must preserve the architecture justification: Independent safety and Ethernet clocks	RFQ
Input Protection	Reverse battery / surge front end	Automotive ideal-diode/eFuse + TVS	Discrete MOSFET + controller	Nomination shall close: ISO16750/7637/OEM transients, low loss, diagnostics. Required evidence: Transient DVP/WCCA. The selected solution must preserve the architecture justification: Protects central ECU LV input	Design
Power	Hold-up energy	Bulk/polymer capacitor bank	Supercap optional if needed	Nomination shall close: 50ms safety-domain hold-up, ESR, temp, lifetime. Required evidence: Energy calculation/aging. The selected solution must preserve the architecture justification: Allows safe-state/log on power collapse	Design
Thermal	Heat spreader/heatsink	Conduction plate / liquid cold plate if platform provides	Forced air heat sink	Nomination shall close: <=75W peak, acoustic, ambient, packaging. Required evidence: CFD/chamber. The selected solution must preserve the architecture justification: Centralized HPA thermal density	Design
Display	Display SerDes/DP bridge	Automotive DP/eDP/FPD-Link/GMSL class	APIX class	Nomination shall close: Display bandwidth, safety monitoring, EMC, cable length. Required evidence: Link/SI/EMC. The selected solution must preserve the architecture justification: Raw pixels should not consume 1G vehicle Ethernet	RFQ
Audio	Audio codec / amp interface	Automotive audio codec/I2S/TDM	SoC integrated	Nomination shall close: Warning audio path and infotainment quality. Required evidence:	RFQ

Domain	Item	Primary/Class	Alternate	Nomination Evidence & Design Rationale	Status
				Audio/HMI DVP. The selected solution must preserve the architecture justification: Cockpit integration	
PCB	High-density multilayer PCB	Automotive HDI 10-16+ layer class	Two-board mezzanine	Nomination shall close: LPDDR/PCIe/Ethernet SI, isolation, thermals. Required evidence: Stack-up/SI/PI/DFM. The selected solution must preserve the architecture justification: Central compute density	Design
Connector	Vehicle I/O connector	Automotive sealed/unsealed connector per location	Supplier alternate	Nomination shall close: Current, pin count, CAN/Ethernet SI, retention, service. Required evidence: LV214/USCAR/OEM tests. The selected solution must preserve the architecture justification: Vehicle interface	RFQ
Connector	High-speed Ethernet connector	H-MTD/MATEnet class	OEM approved alternate	Nomination shall close: 1000/2.5/10GT1 channel, EMC, retention. Required evidence: Channel S-parameter. The selected solution must preserve the architecture justification: High-speed backbone	RFQ
Connector	Display high-speed connector	HFM/Mini-FAKRA class	OEM approved alternate	Nomination shall close: SerDes/DP channel, EMC, vibration. Required evidence: Channel/vibration. The selected solution must preserve the architecture justification: Cockpit displays	RFQ
Diagnostics	DoIP service port/gateway path	Ethernet service interface	Vehicle gateway external port	Nomination shall close: Secure diagnostics, factory/service. Required evidence: DoIP/UDS tests. The selected solution must preserve the architecture justification: Programming/diagnostics	Design
Safety	External supervisor optional	Automotive safety supervisor	PMIC integrated only	Nomination shall close: Diverse watchdog/reset/safe outputs. Required evidence: Safety analysis. The selected solution must preserve the architecture justification: Additional common-cause mitigation if safety case needs	Option
EMC	Common-mode chokes/filtering	Automotive CAN/Ethernet/filter components	No-CMC where SI permits	Nomination shall close: CISPR25/ISO11452 vs SI tradeoff. Required evidence: EMC/SI DVP. The selected solution must preserve the architecture justification: EMC closure	Design
Protection	ESD/TVS array	Automotive low-capacitance protection	Integrated PHY protection	Nomination shall close: ISO10605/OEM ESD, line capacitance. Required evidence: ESD/SI validation. The selected solution must preserve the architecture justification: External pin robustness	Design
Mechanical	ECU enclosure	Die-cast aluminum/EMI sealed	Stamped/hybrid	Nomination shall close: Thermal, EMC shielding, crash mounting, serviceability. Required evidence: Mechanical/EMC/thermal. The selected solution must preserve the architecture justification: Central ECU protection	Design
Manufacturing	Secure provisioning fixture	HSM-backed EOL tool	Offline secure programmer	Nomination shall close: Keys, VIN, variant, calibration, station audit. Required evidence: EOL validation. The selected solution must preserve the architecture justification:	Manufacturing

Domain	Item	Primary/Class	Alternate	Nomination Evidence & Design Rationale	Status
				Production cybersecurity	
Software	OTA client/bootloader	ISO24089/R156-capable update client	OEM middleware	Nomination shall close: Atomic activation, dependency, recovery. Required evidence: Update DVP. The selected solution must preserve the architecture justification: Lifecycle update safety	RFQ
Software	Diagnostics stack	UDS + DoIP stack	Supplier/OEM stack	Nomination shall close: ISO14229/13400 services, security. Required evidence: Conformance. The selected solution must preserve the architecture justification: Service interoperability	RFQ
Software	SOME/IP/ara::com middleware	AUTOSAR Adaptive/SOME-IP class	DDS/OEM service bus	Nomination shall close: Versioned IPC/network services. Required evidence: Integration/load. The selected solution must preserve the architecture justification: Centralized service architecture	RFQ
Software	IDS/firewall	Automotive IDS + gateway firewall	OEM security stack	Nomination shall close: CAN/Ethernet policy, logs, performance. Required evidence: Pen/load test. The selected solution must preserve the architecture justification: CSMS/TARA controls	RFQ
Test	HIL rack	dSPACE SCALEXIO/MicroAutoBox class	NI/Vector HIL class	Nomination shall close: Plant models, fault injection, timing. Required evidence: Bench qualification. The selected solution must preserve the architecture justification: SYS4/SYS5 automation	Program
Test	Network test suite	Vector CANoe/CANalyzer class	ETAS/other	Nomination shall close: CAN FD/Ethernet/DoIP/SOME-IP/SecOC/E2E. Required evidence: Tool qualification as needed. The selected solution must preserve the architecture justification: Network verification	Program
Test	Model-based development	MATLAB/Simulink/ Stateflow + TargetLink class	Embedded Coder/hand code	Nomination shall close: Control modeling, code gen, trace, test. Required evidence: Tool qualification/validation. The selected solution must preserve the architecture justification: Torque/mode/HV algorithms	Program
ALM	Requirements/traceability	IBM DOORS Next / Polarion ALM	Jama/Codebeamer	Nomination shall close: Bidirectional ASPICE RTM/change baselines. Required evidence: Tool config/audit. The selected solution must preserve the architecture justification: Process evidence	Program
Cyber	PKI/backend integration	OEM CSMS/SUMS backend	Supplier cloud stack	Nomination shall close: Certificates, OTA auth, revocation, fleet logs. Required evidence: Pen/audit. The selected solution must preserve the architecture justification: R155/R156/21434 lifecycle	Program
Calibration	Calibration toolchain	CANape/INCA class	OEM calibration tool	Nomination shall close: A2L/calibration governance, signed release. Required evidence: Calibration audit. The selected solution must preserve the architecture justification: Mode/torque/thermal calibration	Program
Service	Secure service tester	DoIP/UDS service tester	OEM diagnostic tester	Nomination shall close: Authenticated routines/update/config. Required evidence:	Program

Domain	Item	Primary/Class	Alternate	Nomination Evidence & Design Rationale	Status
				Service validation. The selected solution must preserve the architecture justification: After-sales support	
Manufacturing	Boundary-scan / ICT	Production ICT/boundary scan	Functional test only	Nomination shall close: PCB shorts/opens/programming . Required evidence: MSA/GRR. The selected solution must preserve the architecture justification: Quality and yield	Manufacturing
Manufacturing	Thermal/functional run-in	Burn-in/run-in station	Sample-based audit	Nomination shall close: Power, thermal, network, memory stress. Required evidence: Quality study. The selected solution must preserve the architecture justification: Early-life defect screening	Manufacturing

Appendix E - Program Work Packages and Gate Acceptance

WP	Work Package	Engineering Scope	Exit / Gate Evidence
WP01	Item definition / legacy migration	Legacy ECU inventory, feature ownership, authority matrix, variant tree, current interfaces and diagnostic ownership.	Approved item boundary, RA baseline, migration inventory and open-interface list.
WP02	Safety & cybersecurity concept	HARA/FSC/TARA, safety/cyber goals, assumptions, threat/safety boundaries and external ECU safety contracts.	Safety/cyber concept review passed; no unresolved authority ambiguity.
WP03	Central hardware architecture	Safety MCU, HPA SoC, PMIC, memory/storage, Ethernet/CAN, display I/O, thermal/mechanical and PCB technology.	CDR with WCCA, initial SI/PI/thermal model and selected A-sample parts.
WP04	Hypervisor / OS / BSW platform	Safety MCU AUTOSAR, Type-1 hypervisor, HPA safety services, Android/Linux/QNX partitions, device assignment and boot chain.	Platform boot, isolation and diagnostic evidence on target hardware.
WP05	Vehicle motion controls	Torque, regen, HV, mode, thermal and degraded-state algorithms with calibrations and independent safety supervision.	SIL/HIL coverage; timing and safety reaction budgets closed.
WP06	Network and middleware	CAN FD DBC/ARXML, Ethernet VLAN/QoS, service contracts, time sync, E2E/SecOC, gateway filtering and diagnostics.	Network schedule, overload/fault evidence and controlled ICD release.
WP07	Cockpit / HMI integration	Driver modes, telltales, warnings, vehicle state, audio/display integration and fallback presentation.	HMI latency, safety telltale and mode usability tests passed.
WP08	Diagnostics / OTA / cybersecurity	UDS/DoIP, secure boot, HSM, update, key lifecycle, logging/IDS, service and plant permissions.	Cybersecurity test report and update recovery evidence.
WP09	DV / HIL / dyno / vehicle	System integration, fault injection, chassis/brake/HV/thermal/network/electrical/EMC verification.	DV gate with no open safety-critical deviation.
WP10	PV / EOL / SOP	Production-intent hardware, EOL fixtures, secure provisioning, MSA, vehicle PV and supplier production readiness.	PV gate / SOP release dossier complete.
WP11	Post-SOP / FRACAS	Field monitoring, DTC/event analysis, software update, supplier quality, cybersecurity monitoring and obsolescence.	Closed-loop field issue management and controlled maintenance releases.

Appendix F - Review and Release Checklist

Check ID	Area	Review Criterion	Gate	Objective Evidence
SAFE-01	Safety & Motion	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.
SAFE-02	Safety & Motion	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.
SAFE-03	Safety & Motion	Verification method, observables and release evidence are defined.	DV/PV readiness review	HIL fault injection, timing trace, local-inverter/ESC response and safety-case evidence.
DRIV-01	Driver Modes	Architecture ownership and safety authority are explicit	Architecture review	Mode-transition HIL and vehicle measurements

Check ID	Area	Review Criterion	Gate	Objective Evidence
		and conflict-free.		covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.
DRIV-02	Driver Modes	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.
DRIV-03	Driver Modes	Verification method, observables and release evidence are defined.	DV/PV readiness review	Mode-transition HIL and vehicle measurements covering jerk, rejected/invalid mode, thermal/SOP clamps and Track/Valet restrictions.
HV C-01	HV Coordination	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.
HV C-02	HV Coordination	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.
HV C-03	HV Coordination	Verification method, observables and release evidence are defined.	DV/PV readiness review	BMS/inverter HIL, welded/failed contactor simulation, precharge timeout, crash/HVIL and shutdown sequence evidence.
THER-01	Thermal & Energy	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.
THER-02	Thermal & Energy	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.
THER-03	Thermal & Energy	Verification method, observables and release evidence are defined.	DV/PV readiness review	Thermal chamber/HIL and dyno testing at cold/hot/SOP boundaries, sensor faults and communication timeouts.
CHAS-01	Chassis & Regen	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.
CHAS-02	Chassis & Regen	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.
CHAS-03	Chassis & Regen	Verification method, observables and release evidence are defined.	DV/PV readiness review	Brake dyno/vehicle blend tests, ABS/ESC activation, low-friction split-mu, BMS charge-SOP loss and local-ECU refusal tests.
CENT-01	Central Compute	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.
CENT-02	Central Compute	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.
CENT-03	Central Compute	Verification method, observables and release evidence are defined.	DV/PV readiness review	WCET/load stress, memory/ECC faults, hypervisor/guest crash, PMIC/brownout and resource-exhaustion fault injection.
NETW-01	Networking	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.
NETW-02	Networking	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and bus-off scenarios.
NETW-03	Networking	Verification method, observables and release evidence are defined.	DV/PV readiness review	Bit-accurate CAN schedule, Ethernet VLAN/QoS/time-sync tests, gateway overload, packet loss, stale/replay and

Check ID	Area	Review Criterion	Gate	Objective Evidence
				bus-off scenarios.
COCK-01	Cockpit/HMI	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.
COCK-02	Cockpit/HMI	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.
COCK-03	Cockpit/HMI	Verification method, observables and release evidence are defined.	DV/PV readiness review	HMI latency, display reset, lost acknowledgement, telltale fallback, physical control and driver-mode usability tests.
DIAG-01	Diagnostics/Service	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.
DIAG-02	Diagnostics/Service	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.
DIAG-03	Diagnostics/Service	Verification method, observables and release evidence are defined.	DV/PV readiness review	UDS/DoIP conformance, session/security negative tests, interrupted routines and plant/service fault-injection tests.
CYBE-01	Cybersecurity	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.
CYBE-02	Cybersecurity	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.
CYBE-03	Cybersecurity	Verification method, observables and release evidence are defined.	DV/PV readiness review	TARA-based penetration, secure boot/update rollback, key/certificate, replay/spoofing and compromised-cockpit tests.
ELEC-01	Electrical/EMC	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.
ELEC-02	Electrical/EMC	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.
ELEC-03	Electrical/EMC	Verification method, observables and release evidence are defined.	DV/PV readiness review	ISO 16750/7637 electrical tests, CISPR 25/ISO 11452 EMC, PI/SI, thermal chamber and load-step measurements.
LIFE-01	Lifecycle/Quality	Architecture ownership and safety authority are explicit and conflict-free.	Architecture review	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.
LIFE-02	Lifecycle/Quality	All limits/calibrations have an owner, source and change-control path.	Requirements/ICD review	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.
LIFE-03	Lifecycle/Quality	Verification method, observables and release evidence are defined.	DV/PV readiness review	EOL station MSA, secure provisioning, PV audits, supplier PCN/EOL reviews, field FRACAS and post-SOP update evidence.

Appendix G - RA Verification Design Matrix

This matrix is deliberately more detailed than a normal concept document because VCU centralization creates system-level coupling across safety, cockpit, network and lifecycle domains. It provides a reusable starting verification design for each requirement authority item; detailed SYS.4/SYS.5 test cases are derived from these rows rather than invented late in DV.

RA ID	Area	Test / Analysis Configuration	Boundary / Fault Stimulus	Observables	Concept-Level Acceptance
RA-VCU-001	Safety & Motion	Vehicle HIL + inverter plant + independent safety monitor	Pedal/brake/gear/ESC faults, HPA crash, stuck/biased requests	Torque request/reason, fault qualification, inhibit time, inverter feedback	No command exceeds bounded capability; unsafe positive torque is revoked inside allocated reaction time.
RA-VCU-002	Driver Modes	Vehicle HIL + HMI + chassis/thermal models	All mode requests during acceleration, braking, thermal/SOP derate, ESC intervention and invalid user/market state	Accepted mode, torque/regen/chassis profile, blend timer, jerk, reason code	Mode never raises authority above capability; acknowledgement and jerk-bounded blend meet released targets.
RA-VCU-003	HV Coordination	BMS/inverter HIL with contactor/precharge plant	Wake/READY, precharge fail, welded feedback, crash/HVIL, BMS reset, inverter not-ready	VCU HV state, BMS command/feedback, DC-link ratio, timeout, torque enable	VCU cannot enable torque before BMS HV-ready/inverter-ready; shutdown is deterministic and non-conflicting.
RA-VCU-004	Thermal & Energy	Thermal HIL/dyno with BMS/inverter/motor capability maps	Cold/hot limits, cooling faults, stale SOP, rapid power demand, recovery hysteresis	Available power/torque, derate reason, temperatures, cooling request, recovery ramp	Capability equals the minimum valid subsystem limit and never increases on invalid/stale input.
RA-VCU-005	Chassis & Regen	Brake/ESC HIL + dyno/vehicle	ABS/ESC active, low-mu, brake press, full SOC, cold battery, inverter regen loss	Requested regen, executed regen, friction command/status, wheel slip, decel/jerk	ESC/brake authority dominates; regen fades safely without wheel lock or unacceptable decel discontinuity.
RA-VCU-006	Central Compute	Target ECU stress + HIL + power/thermal chamber	CPU/GPU/memory overload, guest crash, hypervisor fault injection, ECC/clock/PMIC faults	Safety task WCET, watchdogs, rail/reset, IPC age, torque/HV outputs	Safety partition retains deterministic behavior or transitions safe independent of HPA guest availability.
RA-VCU-007	Networking	CANoe/Ethernet test bench + HIL	Worst-case arbitration, bus-off, packet loss, time-sync loss, gateway overload, replay/stale frames	Bus utilization, WCRT, age/E2E/SecOC status, gateway latency, degraded state	Safety deadlines are met with reserve; invalid/stale/replayed data cannot increase authority.
RA-VCU-008	Cockpit/HMI	Cockpit bench + vehicle HIL	Android/display reset, frozen screen, invalid mode request, telltale source loss	Presentation latency, acknowledgement, fallback indication, safety status	Critical warning/mode status remains correct or fallback path engages; HMI failure does not alter safety authority.
RA-VCU-009	Diagnostics/Service	UDS/DoIP bench + plant/service state model	Unauthorized routine, wrong session, interrupted programming, service while driving	Session/security state, routine permission, DTC/freeze frame, motion/HV state	No diagnostic/service path bypasses motion/HV interlocks; evidence is auditable and recoverable.
RA-VCU-010	Cybersecurity	Cyber HIL / penetration / update rig	Compromised IVI, spoof/replay, debug attempt, key error, rollback, interrupted update	HSM/boot/update verdict, SecOC freshness/auth, gateway decision, security log	Unauthorized code/data/commands are rejected and safety availability/reaction remains bounded.
RA-VCU-011	Electrical/EMC	Electrical transient/EMC/thermal/SI-PI bench	Cold crank, overvoltage, EFT/transient, RF immunity, max HPA/display traffic	Rail droop/reset, emissions/immunity, memory/network errors, safe outputs	No unsafe output; ECU recovers deterministically and meets released EMC/electrical criteria.
RA-VCU-012	Lifecycle/Quality	EOL/PV/service audit + FRACAS workflow	Wrong variant, corrupt calibration, station drift, supplier PCN/EOL, field defect	Genealogy, signature/hash, MSA, release record, change impact, FRACAS closure	Only compatible signed configuration releases; quality/lifecycle evidence remains traceable through support period.
RA-VCU-013	Safety & Motion	Vehicle HIL + inverter plant + independent safety monitor	Pedal/brake/gear/ESC faults, HPA crash, stuck/biased requests	Torque request/reason, fault qualification, inhibit time, inverter feedback	No command exceeds bounded capability; unsafe positive torque is revoked inside allocated reaction time.
RA-VCU-014	Driver Modes	Vehicle HIL + HMI + chassis/thermal models	All mode requests during acceleration, braking, thermal/SOP derate, ESC intervention and invalid user/market state	Accepted mode, torque/regen/chassis profile, blend timer, jerk, reason code	Mode never raises authority above capability; acknowledgement and jerk-bounded blend meet released targets.
RA-VCU-015	HV Coordination	BMS/inverter HIL with contactor/precharge plant	Wake/READY, precharge fail, welded feedback, crash/HVIL, BMS reset, inverter not-ready	VCU HV state, BMS command/feedback, DC-link ratio, timeout, torque enable	VCU cannot enable torque before BMS HV-ready/inverter-ready; shutdown is deterministic and non-conflicting.
RA-VCU-016	Thermal & Energy	Thermal HIL/dyno with BMS/inverter/motor capability maps	Cold/hot limits, cooling faults, stale SOP, rapid power demand, recovery hysteresis	Available power/torque, derate reason, temperatures, cooling request, recovery ramp	Capability equals the minimum valid subsystem limit and never increases on invalid/stale input.
RA-VCU-017	Chassis & Regen	Brake/ESC HIL + dyno/vehicle	ABS/ESC active, low-mu, brake press, full SOC, cold battery, inverter regen loss	Requested regen, executed regen, friction command/status, wheel slip, decel/jerk	ESC/brake authority dominates; regen fades safely without wheel lock or unacceptable decel discontinuity.
RA-VCU-018	Central Compute	Target ECU stress + HIL +	CPU/GPU/memory	Safety task WCET,	Safety partition retains

RA ID	Area	Test / Analysis Configuration	Boundary / Fault Stimulus	Observables	Concept-Level Acceptance
		power/thermal chamber	overload, guest crash, hypervisor fault injection, ECC/clock/PMIC faults	watchdogs, rail/reset, IPC age, torque/HV outputs	deterministic behavior or transitions safe independent of HPA guest availability.
RA-VCU-019	Networking	CANoe/Ethernet test bench + HIL	Worst-case arbitration, bus-off, packet loss, time-sync loss, gateway overload, replay/stale frames	Bus utilization, WCRT, age/E2E/SecOC status, gateway latency, degraded state	Safety deadlines are met with reserve; invalid/stale/replayed data cannot increase authority.
RA-VCU-020	Cockpit/HMI	Cockpit bench + vehicle HIL	Android/display reset, frozen screen, invalid mode request, telltale source loss	Presentation latency, acknowledgement, fallback indication, safety status	Critical warning/mode status remains correct or fallback path engages; HMI failure does not alter safety authority.
RA-VCU-021	Diagnostics/Service	UDS/DoIP bench + plant/service state model	Unauthorized routine, wrong session, interrupted programming, service while driving	Session/security state, routine permission, DTC/freeze frame, motion/HV state	No diagnostic/service path bypasses motion/HV interlocks; evidence is auditable and recoverable.
RA-VCU-022	Cybersecurity	Cyber HIL / penetration / update rig	Compromised IVI, spoof/replay, debug attempt, key error, rollback, interrupted update	HSM/boot/update verdict, SecOC freshness/auth, gateway decision, security log	Unauthorized code/data/commands are rejected and safety availability/reaction remains bounded.
RA-VCU-023	Electrical/EMC	Electrical transient/EMC/thermal/SI-PI bench	Cold crank, overvoltage, EFT/transient, RF immunity, max HPA/display traffic	Rail droop/reset, emissions/immunity, memory/network errors, safe outputs	No unsafe output; ECU recovers deterministically and meets released EMC/electrical criteria.
RA-VCU-024	Lifecycle/Quality	EOL/PV/service audit + FRACAS workflow	Wrong variant, corrupt calibration, station drift, supplier PCN/EOL, field defect	Genealogy, signature/hash, MSA, release record, change impact, FRACAS closure	Only compatible signed configuration releases; quality/lifecycle evidence remains traceable through support period.
RA-VCU-025	Safety & Motion	Vehicle HIL + inverter plant + independent safety monitor	Pedal/brake/gear/ESC faults, HPA crash, stuck/biased requests	Torque request/reason, fault qualification, inhibit time, inverter feedback	No command exceeds bounded capability; unsafe positive torque is revoked inside allocated reaction time.
RA-VCU-026	Driver Modes	Vehicle HIL + HMI + chassis/thermal models	All mode requests during acceleration, braking, thermal/SOP derate, ESC intervention and invalid user/market state	Accepted mode, torque/regen/chassis profile, blend timer, jerk, reason code	Mode never raises authority above capability; acknowledgement and jerk-bounded blend meet released targets.
RA-VCU-027	HV Coordination	BMS/inverter HIL with contactor/precharge plant	Wake/READY, precharge fail, welded feedback, crash/HVIL, BMS reset, inverter not-ready	VCU HV state, BMS command/feedback, DC-link ratio, timeout, torque enable	VCU cannot enable torque before BMS HV-ready/inverter-ready; shutdown is deterministic and non-conflicting.
RA-VCU-028	Thermal & Energy	Thermal HIL/dyno with BMS/inverter/motor capability maps	Cold/hot limits, cooling faults, stale SOP, rapid power demand, recovery hysteresis	Available power/torque, derate reason, temperatures, cooling request, recovery ramp	Capability equals the minimum valid subsystem limit and never increases on invalid/stale input.
RA-VCU-029	Chassis & Regen	Brake/ESC HIL + dyno/vehicle	ABS/ESC active, low-mu, brake press, full SOC, cold battery, inverter regen loss	Requested regen, executed regen, friction command/status, wheel slip, decel/jerk	ESC/brake authority dominates; regen fades safely without wheel lock or unacceptable decel discontinuity.
RA-VCU-030	Central Compute	Target ECU stress + HIL + power/thermal chamber	CPU/GPU/memory overload, guest crash, hypervisor fault injection, ECC/clock/PMIC faults	Safety task WCET, watchdogs, rail/reset, IPC age, torque/HV outputs	Safety partition retains deterministic behavior or transitions safe independent of HPA guest availability.
RA-VCU-031	Networking	CANoe/Ethernet test bench + HIL	Worst-case arbitration, bus-off, packet loss, time-sync loss, gateway overload, replay/stale frames	Bus utilization, WCRT, age/E2E/SecOC status, gateway latency, degraded state	Safety deadlines are met with reserve; invalid/stale/replayed data cannot increase authority.
RA-VCU-032	Cockpit/HMI	Cockpit bench + vehicle HIL	Android/display reset, frozen screen, invalid mode request, telltale source loss	Presentation latency, acknowledgement, fallback indication, safety status	Critical warning/mode status remains correct or fallback path engages; HMI failure does not alter safety authority.
RA-VCU-033	Diagnostics/Service	UDS/DoIP bench + plant/service state model	Unauthorized routine, wrong session, interrupted programming, service while driving	Session/security state, routine permission, DTC/freeze frame, motion/HV state	No diagnostic/service path bypasses motion/HV interlocks; evidence is auditable and recoverable.
RA-VCU-034	Cybersecurity	Cyber HIL / penetration / update rig	Compromised IVI, spoof/replay, debug attempt, key error, rollback, interrupted update	HSM/boot/update verdict, SecOC freshness/auth, gateway decision, security log	Unauthorized code/data/commands are rejected and safety availability/reaction remains bounded.
RA-VCU-035	Electrical/EMC	Electrical transient/EMC/thermal/SI-PI bench	Cold crank, overvoltage, EFT/transient, RF immunity, max HPA/display traffic	Rail droop/reset, emissions/immunity, memory/network errors, safe outputs	No unsafe output; ECU recovers deterministically and meets released EMC/electrical criteria.
RA-VCU-036	Lifecycle/Quality	EOL/PV/service audit + FRACAS workflow	Wrong variant, corrupt calibration, station drift, supplier PCN/EOL, field defect	Genealogy, signature/hash, MSA, release record, change impact, FRACAS closure	Only compatible signed configuration releases; quality/lifecycle evidence remains traceable through support period.
RA-VCU-037	Safety & Motion	Vehicle HIL + inverter	Pedal/brake/gear/ESC	Torque request/reason,	No command exceeds

RA ID	Area	Test / Analysis Configuration	Boundary / Fault Stimulus	Observables	Concept-Level Acceptance
		plant + independent safety monitor	faults, HPA crash, stuck/biased requests	fault qualification, inhibit time, inverter feedback	bounded capability; unsafe positive torque is revoked inside allocated reaction time.
RA-VCU-038	Driver Modes	Vehicle HIL + HMI + chassis/thermal models	All mode requests during acceleration, braking, thermal/SOP derate, ESC intervention and invalid user/market state	Accepted mode, torque/regen/chassis profile, blend timer, jerk, reason code	Mode never raises authority above capability; acknowledgement and jerk-bounded blend meet released targets.
RA-VCU-039	HV Coordination	BMS/inverter HIL with contactor/precharge plant	Wake/READY, precharge fail, welded feedback, crash/HVIL, BMS reset, inverter not-ready	VCU HV state, BMS command/feedback, DC-link ratio, timeout, torque enable	VCU cannot enable torque before BMS HV-ready/inverter-ready; shutdown is deterministic and non-conflicting.
RA-VCU-040	Thermal & Energy	Thermal HIL/dyno with BMS/inverter/motor capability maps	Cold/hot limits, cooling faults, stale SOP, rapid power demand, recovery hysteresis	Available power/torque, derate reason, temperatures, cooling request, recovery ramp	Capability equals the minimum valid subsystem limit and never increases on invalid/stale input.
RA-VCU-041	Chassis & Regen	Brake/ESC HIL + dyno/vehicle	ABS/ESC active, low-mu, brake press, full SOC, cold battery, inverter regen loss	Requested regen, executed regen, friction command/status, wheel slip, decel/jerk	ESC/brake authority dominates; regen fades safely without wheel lock or unacceptable decel discontinuity.
RA-VCU-042	Central Compute	Target ECU stress + HIL + power/thermal chamber	CPU/GPU/memory overload, guest crash, hypervisor fault injection, ECC/clock/PMIC faults	Safety task WCET, watchdogs, rail/reset, IPC age, torque/HV outputs	Safety partition retains deterministic behavior or transitions safe independent of HPA guest availability.
RA-VCU-043	Networking	CANoe/Ethernet test bench + HIL	Worst-case arbitration, bus-off, packet loss, time-sync loss, gateway overload, replay/stale frames	Bus utilization, WCRT, age/E2E/SecOC status, gateway latency, degraded state	Safety deadlines are met with reserve; invalid/stale/replayed data cannot increase authority.
RA-VCU-044	Cockpit/HMI	Cockpit bench + vehicle HIL	Android/display reset, frozen screen, invalid mode request, telltale source loss	Presentation latency, acknowledgement, fallback indication, safety status	Critical warning/mode status remains correct or fallback path engages; HMI failure does not alter safety authority.
RA-VCU-045	Diagnostics/Service	UDS/DoIP bench + plant/service state model	Unauthorized routine, wrong session, interrupted programming, service while driving	Session/security state, routine permission, DTC/freeze frame, motion/HV state	No diagnostic/service path bypasses motion/HV interlocks; evidence is auditable and recoverable.
RA-VCU-046	Cybersecurity	Cyber HIL / penetration / update rig	Compromised IVI, spoof/replay, debug attempt, key error, rollback, interrupted update	HSM/boot/update verdict, SecOC freshness/auth, gateway decision, security log	Unauthorized code/data/commands are rejected and safety availability/reaction remains bounded.
RA-VCU-047	Electrical/EMC	Electrical transient/EMC/thermal/SI-PI bench	Cold crank, overvoltage, EFT/transient, RF immunity, max HPA/display traffic	Rail droop/reset, emissions/immunity, memory/network errors, safe outputs	No unsafe output; ECU recovers deterministically and meets released EMC/electrical criteria.
RA-VCU-048	Lifecycle/Quality	EOL/PV/service audit + FRACAS workflow	Wrong variant, corrupt calibration, station drift, supplier PCN/EOL, field defect	Genealogy, signature/hash, MSA, release record, change impact, FRACAS closure	Only compatible signed configuration releases; quality/lifecycle evidence remains traceable through support period.
RA-VCU-049	Safety & Motion	Vehicle HIL + inverter plant + independent safety monitor	Pedal/brake/gear/ESC faults, HPA crash, stuck/biased requests	Torque request/reason, fault qualification, inhibit time, inverter feedback	No command exceeds bounded capability; unsafe positive torque is revoked inside allocated reaction time.
RA-VCU-050	Driver Modes	Vehicle HIL + HMI + chassis/thermal models	All mode requests during acceleration, braking, thermal/SOP derate, ESC intervention and invalid user/market state	Accepted mode, torque/regen/chassis profile, blend timer, jerk, reason code	Mode never raises authority above capability; acknowledgement and jerk-bounded blend meet released targets.
RA-VCU-051	HV Coordination	BMS/inverter HIL with contactor/precharge plant	Wake/READY, precharge fail, welded feedback, crash/HVIL, BMS reset, inverter not-ready	VCU HV state, BMS command/feedback, DC-link ratio, timeout, torque enable	VCU cannot enable torque before BMS HV-ready/inverter-ready; shutdown is deterministic and non-conflicting.
RA-VCU-052	Thermal & Energy	Thermal HIL/dyno with BMS/inverter/motor capability maps	Cold/hot limits, cooling faults, stale SOP, rapid power demand, recovery hysteresis	Available power/torque, derate reason, temperatures, cooling request, recovery ramp	Capability equals the minimum valid subsystem limit and never increases on invalid/stale input.
RA-VCU-053	Chassis & Regen	Brake/ESC HIL + dyno/vehicle	ABS/ESC active, low-mu, brake press, full SOC, cold battery, inverter regen loss	Requested regen, executed regen, friction command/status, wheel slip, decel/jerk	ESC/brake authority dominates; regen fades safely without wheel lock or unacceptable decel discontinuity.
RA-VCU-054	Central Compute	Target ECU stress + HIL + power/thermal chamber	CPU/GPU/memory overload, guest crash, hypervisor fault injection, ECC/clock/PMIC faults	Safety task WCET, watchdogs, rail/reset, IPC age, torque/HV outputs	Safety partition retains deterministic behavior or transitions safe independent of HPA guest availability.
RA-VCU-055	Networking	CANoe/Ethernet test bench + HIL	Worst-case arbitration, bus-off, packet loss, time-sync loss, gateway overload, replay/stale frames	Bus utilization, WCRT, age/E2E/SecOC status, gateway latency, degraded state	Safety deadlines are met with reserve; invalid/stale/replayed data cannot increase authority.

RA ID	Area	Test / Analysis Configuration	Boundary / Fault Stimulus	Observables	Concept-Level Acceptance
RA-VCU-056	Cockpit/HMI	Cockpit bench + vehicle HIL	Android/display reset, frozen screen, invalid mode request, telltale source loss	Presentation latency, acknowledgement, fallback indication, safety status	Critical warning/mode status remains correct or fallback path engages; HMI failure does not alter safety authority.
RA-VCU-057	Diagnostics/Service	UDS/DoIP bench + plant/service state model	Unauthorized routine, wrong session, interrupted programming, service while driving	Session/security state, routine permission, DTC/freeze frame, motion/HV state	No diagnostic/service path bypasses motion/HV interlocks; evidence is auditable and recoverable.
RA-VCU-058	Cybersecurity	Cyber HIL / penetration / update rig	Compromised IVI, spoof/replay, debug attempt, key error, rollback, interrupted update	HSM/boot/update verdict, SecOC freshness/auth, gateway decision, security log	Unauthorized code/data/commands are rejected and safety availability/reaction remains bounded.
RA-VCU-059	Electrical/EMC	Electrical transient/EMC/thermal/SI-PI bench	Cold crank, overvoltage, EFT/transient, RF immunity, max HPA/display traffic	Rail droop/reset, emissions/immunity, memory/network errors, safe outputs	No unsafe output; ECU recovers deterministically and meets released EMC/electrical criteria.
RA-VCU-060	Lifecycle/Quality	EOL/PV/service audit + FRACAS workflow	Wrong variant, corrupt calibration, station drift, supplier PCN/EOL, field defect	Genealogy, signature/hash, MSA, release record, change impact, FRACAS closure	Only compatible signed configuration releases; quality/lifecycle evidence remains traceable through support period.

Appendix H - Supplier / Component Nomination Decision Matrix

Component nomination must demonstrate system suitability, not simply datasheet feature compliance. Each BOM line is therefore reviewed against functional fit, safety/cyber evidence, integration evidence, lifecycle/supply evidence and verification readiness before B-sample freeze.

Domain	Item	Primary/Class	Alternate	Functional / Integration Gate	Safety / Cyber Gate	Lifecycle / Supplier Gate	Status
Compute	Safety MCU	Infineon AURIX TC4x class	Renesas RH850/U2B / NXP S32N class	Functional: Lockstep, ECC, HSM, CAN/Ethernet, safety manual, AUTOSAR ecosystem, lifecycle; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Safety manual, FMEDA, MCU benchmark, PCN/EOL; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Compute	Application/Cockpit SoC	Renesas R-Car X5H class	Qualcomm Snapdragon Cockpit Elite class	Functional: CPU/GPU, virtualization, multi-display, memory BW, power, safety island, Android/Linux/QNX support, lifecycle; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Target workload benchmark, thermal, safety/security collateral; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Compute	Safety-centric central processor alternate	NXP S32N55 class	AURIX TC4x + gateway FPGA/ASIC	Functional: Deterministic real-time consolidation, hardware isolation, Ethernet switch/security; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: NXP safety/performance evidence; supplier PCN/EOL and automotive quality evidence required before nomination.	Trade study
Software	Safety hypervisor	QNX Hypervisor for Safety class	Green Hills INTEGRITY Multivisor class / supplier safety hypervisor	Functional: ISO26262 evidence, guest isolation, Android/Linux support, device assignment, boot time; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Safety certificate/manual, latency, supported SoC; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Software	Safety OS/AUTOSAR Classic	AUTOSAR Classic OS/BSW + safety vendor stack	SafeRTOS / proprietary safety RTOS as project architecture permits	Functional: Timing, MCAL, safety qualification, diagnostics, SecOC/E2E; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Safety qualification/toolchain evidence; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Software	Cockpit OS	Android Automotive + Linux/QNX	QNX cockpit / AGL	Functional: OEM UX ecosystem, app	Safety/Cyber: safety manual/FMEDA	Lifecycle: BSP, update, security	RFQ

Domain	Item	Primary/Class	Alternate	Functional / Integration Gate	Safety / Cyber Gate	Lifecycle / Supplier Gate	Status
		services		model, boot time, security, lifecycle; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	assumptions, HSM/secure lifecycle and failure behavior where applicable.	support; supplier PCN/EOL and automotive quality evidence required before nomination.	
Power	Safety PMIC	Infineon TLF35585 class	NXP automotive safety PMIC class	Functional: 3-40V input, watchdog, rail monitoring, safe-state outputs, ASIL support; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: PMIC safety manual/WCCA; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Power	Application SoC PMIC	SoC-vendor PMIC/reference power tree	Automotive multiphase PMIC set	Functional: Transient response, sequencing, efficiency, safety monitors, low-power state; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Vendor reference design/PI model; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Memory	LPDDR5/LPDDR5X	Automotive LPDDR5X	Automotive LPDDR5	Functional: Bandwidth, ECC/path support, temp, lifecycle; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Memory vendor SI/model/qualification; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Storage	UFS / NVMe	Automotive UFS 3.x/4.x or NVMe	Automotive eMMC + secondary storage	Functional: A/B OTA, logs, endurance, power-loss behavior; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Endurance, temp, power-cut qualification; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Memory	Safety NVM	MCU internal flash + external safety-grade NVM optional	FRAM/MRAM class	Functional: Endurance, retention, ECC, power loss; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Qualification/endurance; supplier PCN/EOL and automotive quality evidence required before nomination.	Design
Network	Central Ethernet switch	Marvell Brightlane Q622x class	NXP/other automotive TSN switch	Functional: 1/2.5/10G, TSN, MACsec, VLAN, QoS, diagnostics; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Switch datasheet, thermal, TSN/MACsec; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Network	100/1000BASE-T1 PHY	Brightlane 88Q2112 class	Other OPEN Alliance automotive PHY	Functional: 100/1000T1, TC10, 802.1AS, fail-safe, temp; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: PHY/channel mask/EMC; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Network	CAN FD transceiver	TI TCAN1043A/HG-Q1 class	NXP TJA146x class	Functional: CAN FD, wake/INH, bus fault, low-power, CAN SIC if needed; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: ISO11898 compliance, EMC; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Network	LIN transceiver	Automotive LIN class	Integrated SBC	Functional: Wake, low-power, local console; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: EMC/network test; supplier PCN/EOL and automotive quality evidence required before nomination.	Optional
Security	HSM / secure element	MCU/SoC integrated HSM	External automotive secure element/TPM	Functional: Secure boot, key storage, monotonic counters, crypto throughput; Integration: target benchmark/board bring-up/SI-PI or network evidence as	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Security target/certification; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ

Domain	Item	Primary/Class	Alternate	Functional / Integration Gate	Safety / Cyber Gate	Lifecycle / Supplier Gate	Status
				applicable.			
Clock	Automotive oscillator / RTC	Low-ppm automotive oscillator	TCXO class	Functional: Jitter, temp drift, startup, safety clock independence; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: AEC-Q100 / timing analysis; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Input Protection	Reverse battery / surge front end	Automotive ideal-diode/eFuse + TVS	Discrete MOSFET + controller	Functional: ISO16750/7637/OEM transients, low loss, diagnostics; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Transient DVP/WCCA; supplier PCN/EOL and automotive quality evidence required before nomination.	Design
Power	Hold-up energy	Bulk/polymer capacitor bank	Supercap optional if needed	Functional: 50ms safety-domain hold-up, ESR, temp, lifetime; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Energy calculation/aging; supplier PCN/EOL and automotive quality evidence required before nomination.	Design
Thermal	Heat spreader/heatsink	Conduction plate / liquid cold plate if platform provides	Forced air heat sink	Functional: <=75W peak, acoustic, ambient, packaging; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: CFD/chamber; supplier PCN/EOL and automotive quality evidence required before nomination.	Design
Display	Display SerDes/DP bridge	Automotive DP/eDP/FPD-Link/GMSL class	APIX class	Functional: Display bandwidth, safety monitoring, EMC, cable length; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Link/SI/EMC; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Audio	Audio codec / amp interface	Automotive audio codec/I2S/TDM	SoC integrated	Functional: Warning audio path and infotainment quality; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Audio/HMI DVP; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
PCB	High-density multilayer PCB	Automotive HDI 10-16+ layer class	Two-board mezzanine	Functional: LPDDR/PCIe/Ethernet SI, isolation, thermals; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Stack-up/SI/PI/DFM; supplier PCN/EOL and automotive quality evidence required before nomination.	Design
Connector	Vehicle I/O connector	Automotive sealed/unsealed connector per location	Supplier alternate	Functional: Current, pin count, CAN/Ethernet SI, retention, service; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: LV214/USCAR/OEM tests; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Connector	High-speed Ethernet connector	H-MTD/MATEnet class	OEM approved alternate	Functional: 1000/2.5/10GT1 channel, EMC, retention; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Channel S-parameter; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Connector	Display high-speed connector	HFM/Mini-FAKRA class	OEM approved alternate	Functional: SerDes/DP channel, EMC, vibration; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Channel/vibration; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Diagnostics	DoIP service port/gateway path	Ethernet service interface	Vehicle gateway external port	Functional: Secure diagnostics, factory/service; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: DoIP/UDS tests; supplier PCN/EOL and automotive quality evidence required before nomination.	Design
Safety	External supervisor optional	Automotive safety supervisor	PMIC integrated only	Functional: Diverse watchdog/reset/safe	Safety/Cyber: safety manual/FMEDA	Lifecycle: Safety analysis; supplier	Option

Domain	Item	Primary/Class	Alternate	Functional / Integration Gate	Safety / Cyber Gate	Lifecycle / Supplier Gate	Status
				outputs; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	assumptions, HSM/secure lifecycle and failure behavior where applicable.	PCN/EOL and automotive quality evidence required before nomination.	
EMC	Common-mode chokes/filtering	Automotive CAN/Ethernet/filter components	No-CMC where SI permits	Functional: CISPR25/ISO11452 vs SI tradeoff; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: EMC/SI DVP; supplier PCN/EOL and automotive quality evidence required before nomination.	Design
Protection	ESD/TVS array	Automotive low-capacitance protection	Integrated PHY protection	Functional: ISO10605/OEM ESD, line capacitance; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: ESD/SI validation; supplier PCN/EOL and automotive quality evidence required before nomination.	Design
Mechanical	ECU enclosure	Die-cast aluminum/EMI sealed	Stamped/hybrid	Functional: Thermal, EMC shielding, crash mounting, serviceability; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Mechanical/EMC/thermal; supplier PCN/EOL and automotive quality evidence required before nomination.	Design
Manufacturing	Secure provisioning fixture	HSM-backed EOL tool	Offline secure programmer	Functional: Keys, VIN, variant, calibration, station audit; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: EOL validation; supplier PCN/EOL and automotive quality evidence required before nomination.	Manufacturing
Software	OTA client/bootloader	ISO24089/R156-capable update client	OEM middleware	Functional: Atomic activation, dependency, recovery; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Update DVP; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Software	Diagnostics stack	UDS + DoIP stack	Supplier/OEM stack	Functional: ISO14229/13400 services, security; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Conformance; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Software	SOME/IP/ara::com middleware	AUTOSAR Adaptive/SOME-IP class	DDS/OEM service bus	Functional: Versioned IPC/network services; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Integration/load; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Software	IDS/firewall	Automotive IDS + gateway firewall	OEM security stack	Functional: CAN/Ethernet policy, logs, performance; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Pen/load test; supplier PCN/EOL and automotive quality evidence required before nomination.	RFQ
Test	HIL rack	dSPACE SCALEXIO/MicroAutoBox class	NI/Vector HIL class	Functional: Plant models, fault injection, timing; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Bench qualification; supplier PCN/EOL and automotive quality evidence required before nomination.	Program
Test	Network test suite	Vector CANoe/CANalyzer class	ETAS/other	Functional: CAN FD/Ethernet/DoIP/SOME-IP/SecOC/E2E; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Tool qualification as needed; supplier PCN/EOL and automotive quality evidence required before nomination.	Program
Test	Model-based development	MATLAB/Simulink/Stateflow + TargetLink class	Embedded Coder/hand code	Functional: Control modeling, code gen, trace, test; Integration: target benchmark/board bring-up/SI-PI or network evidence as	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Tool qualification/validation; supplier PCN/EOL and automotive quality evidence required before nomination.	Program

Domain	Item	Primary/Class	Alternate	Functional / Integration Gate	Safety / Cyber Gate	Lifecycle / Supplier Gate	Status
				applicable.			
ALM	Requirements/ traceability	IBM DOORS Next / Polarion ALM	Jama/Codebeamer	Functional: Bidirectional ASPICE RTM/change baselines; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Tool config/audit; supplier PCN/EOL and automotive quality evidence required before nomination.	Program
Cyber	PKI/backend integration	OEM CSMS/SUMS backend	Supplier cloud stack	Functional: Certificates, OTA auth, revocation, fleet logs; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Pen/audit; supplier PCN/EOL and automotive quality evidence required before nomination.	Program
Calibration	Calibration toolchain	CANape/INCA class	OEM calibration tool	Functional: A2L/calibration governance, signed release; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Calibration audit; supplier PCN/EOL and automotive quality evidence required before nomination.	Program
Service	Secure service tester	DoIP/UDS service tester	OEM diagnostic tester	Functional: Authenticated routines/update/config; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Service validation; supplier PCN/EOL and automotive quality evidence required before nomination.	Program
Manufacturing	Boundary-scan / ICT	Production ICT/boundary scan	Functional test only	Functional: PCB shorts/opens/programming; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: MSA/GRR; supplier PCN/EOL and automotive quality evidence required before nomination.	Manufacturing
Manufacturing	Thermal/functional run-in	Burn-in/run-in station	Sample-based audit	Functional: Power, thermal, network, memory stress; Integration: target benchmark/board bring-up/SI-PI or network evidence as applicable.	Safety/Cyber: safety manual/FMEDA assumptions, HSM/secure lifecycle and failure behavior where applicable.	Lifecycle: Quality study; supplier PCN/EOL and automotive quality evidence required before nomination.	Manufacturing

Appendix I - Sample A / B / C Maturity Definition

Stage	Hardware / Software Intent	Required Evidence	Release Boundary
Sample A - Bring-up	Engineering PCB/enclosure with representative MCU/SoC/PMIC/network; interfaces may use debug fixtures.	Boot/power/network/MCAL/BSP/ hypervisor, basic torque/HV I/O stubs, thermal characterization and silicon errata discovery.	No production-representative EMC/mechanical claim; safety evidence is feasibility only.
Sample B - Integrated DV	Vehicle-intent architecture, connectors, thermal solution and software partitions with representative external ECU interfaces.	Complete torque/regen/HV/mode/thermal flows, safety fault injection, cybersecurity baseline, pre-compliance EMC/electrical and network stress.	Architecture and BOM changes after B freeze require formal change impact.
Sample C - Production Intent	Production-intent PCB/BOM/enclosure, secure provisioning, manufacturing interfaces and release-candidate software.	DV/PV closure, EOL fixture correlation, supplier production evidence, vehicle robustness, update/service and homologation support.	Only controlled deviations allowed; open safety-critical issues block release.

Appendix J - Block-by-Block Concept Planning Checklist

This appendix defines the concept-phase planning package expected before each block is considered ready for detailed SYS.3/HWE.1/SWE.1 design. It is intentionally reusable across processor, vehicle-voltage and cockpit variants.

Block	Architecture Scope	Required Inputs / Dependencies	Expected Design Outputs	Freeze Criteria	Required Evidence
B01	LV Power Input, PMIC, Wake/Sleep & Power-State Control	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include 12V input envelope, Power consumption, Safety power domain, Brownout handling.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B02	Independent Safety MCU, Watchdog, Safe-State I/O & Actuation Gateway	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Safety MCU capability, Torque authority, Safe outputs, Cycle time.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B03	Central Application SoC, Memory, Storage, Hypervisor & Hardware Virtualization	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Application SoC headroom, Virtualization, Safety island use, Memory bandwidth.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B04	Vehicle I/O Acquisition, Driver Controls & Discrete Safety Inputs	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Pedal sensing, Brake input, Gear/state input, Ignition/start.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B05	Vehicle State Estimator, Torque Arbitration & Propulsion Request Manager	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Torque equation, Pedal map, Torque clamp, Brake override.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B06	Regenerative Braking, Deceleration Coordination & Brake-Energy Manager	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Decel request, Charge SOP, Wheel slip, Low speed fade.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B07	Multi-Driver Mode Manager, Calibration, Profile Security & Transition Control	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Mode set, Request validation, Ack timing, Blend timing.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B08	Vehicle Dynamics, EPS/ESC/Suspension Coordination & Chassis State Manager	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include ESC authority, EPS mode request, Suspension request, Yaw/slip state.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B09	HV State, Contactor/Precharge Coordination & Charge/Drive Interlocks	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include HV state machine, BMS authority, Precharge, Timeout.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B10	Energy, Thermal,	Inputs/Dependencies:	Design Outputs: block	Freeze Criteria: no	Supplier / Program Evidence:

Block	Architecture Scope	Required Inputs / Dependencies	Expected Design Outputs	Freeze Criteria	Required Evidence
	Performance Limit & Derating Supervisor	controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Limit arbitration, No universal 65C, Derate hysteresis, Peak timer.	context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B11	CAN FD / Automotive Ethernet Gateway, TSN Switching & Network Management	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include CAN architecture, CAN rates, Bus load, Ethernet baseline.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B12	Time Synchronization, Service-Oriented Middleware, IPC & Data Freshness	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Global time, PTP, Safety time, SOME/IP.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B13	Cockpit/IVI/HMI Domain, Safety Telltales & Driver-Interaction Boundary	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Android boundary, Mode request, Safety telltale, Display latency.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B14	Diagnostics, UDS/DoIP, Event Recorder, Service & Fleet Logging	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include UDS services, DoIP, DTC ownership, Freeze frame.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B15	Boot, OTA Software Update, Configuration, Variant & Calibration Management	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Secure boot, A/B update, Safety MCU update, Dependency manifest.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B16	Cybersecurity, HSM, SecOC/MACsec, IDS & Access-Control Services	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include HSM, SecOC, MACsec, IDS.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B17	Functional Safety Supervisor, Freedom-from-Interference & Health Management	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Safety supervisor, FFI, Reaction budget, Safe-state selection.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B18	Central ECU Thermal, EMC, SI/PI, Mechanical & Environmental Design	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Thermal resistance, Power integrity, DDR SI, PCIe/storage SI.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.
B19	Manufacturing, EOL Calibration, Secure Provisioning & Vehicle Commissioning	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Secure provisioning, MCU test, SoC test, Power test.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.

Block	Architecture Scope	Required Inputs / Dependencies	Expected Design Outputs	Freeze Criteria	Required Evidence
			criteria.		
B20	Lifecycle Quality, Reliability, Obsolescence, Field Monitoring & FRACAS	Inputs/Dependencies: controlled SYS.2 allocation, external ICDs, selected component safety/security manuals and vehicle variant assumptions. Key early SYS.3 topics include Mission life, PCN/EOL, FRACAS, Telemetry.	Design Outputs: block context, HW/SW ownership, failure containment, timing/power/network budgets, diagnostics, calibration/configuration ownership, supplier assumptions and measurable acceptance criteria.	Freeze Criteria: no unresolved authority conflict; WCCA/resource budget has margin; safety/cyber assumptions are traceable; interface timeout/degraded behavior is explicit; DVP method and observables are defined.	Supplier / Program Evidence: datasheet/reference manual, safety manual/FMEDA if credited, MCAL/BSP/driver maturity, PCN/EOL support, target benchmark and A/B-sample fault/thermal/network evidence.

Appendix K - Verification Infrastructure and Test Asset Plan

A central VCU/cockpit ECU cannot be validated by a single HIL rack. The program requires a layered evidence environment that separates deterministic motion-control testing, HPA/cockpit stress, network/security testing, physical electrical/EMC validation and vehicle-level behavior.

Asset ID	Test Asset	Reference Environment	Primary Evidence / Purpose	Owner
TA-01	Requirements/ALM environment	DOORS Next / Polarion class	RA/SYS1/SYS2/SYS3/HSR/ SWSR/ICD/RTM baselining, reviews, change history	Configuration admin + Systems
TA-02	MIL/SIL controls environment	MATLAB/Simulink/Stateflow/ TargetLink or equivalent	Torque, regen, mode, HV and thermal plant/control simulation with reproducible model baselines	Controls
TA-03	Safety MCU target bench	AURIX/S32N/RH850-class target + PMIC + transceivers	MCAL/BSW, watchdog, reset, ECC, safety I/O, timing and fault injection	HW/Safety SW
TA-04	HPA SoC bring-up bench	Selected SoC EVK/central-compute board	BSP, hypervisor, Android/Linux/QNX, memory/storage, device assignment and thermal load	Platform
TA-05	Power transient bench	Programmable LV source + electronic loads + scopes	Cold crank, dropouts, overvoltage, load steps, rail sequencing and reset/safe-state	HW/EE
TA-06	CAN FD network bench	CANoe/CANalyzer + restbus + fault insertion	DBC/ARXML, WCRT, bus-off, gateway, E2E/SecOC and timeouts	Network
TA-07	Ethernet/TSN bench	Traffic generator/analyzer + automotive PHY harness	VLAN/QoS/TSN/time sync/MACsec/DoIP/overload and switch/PHY diagnostics	Network
TA-08	Vehicle HIL	dSPACE SCALEXIO/MicroAutoBox or equivalent	Integrated inverter/BMS/ESC/EPS/thermal vehicle plant, motion/HV fault injection	SYS.4/SYS.5
TA-09	Dyno / powertrain bench	Motor/inverter dyno with BMS/vehicle simulation	Torque step, SOP/thermal clamp, regen, efficiency and degraded drive	Controls/Vehicle
TA-10	Brake/chassis bench	Brake/ESC HIL or vehicle dynamics rig	Regen/friction blend, ABS/ESC authority, low-mu and stability interactions	Chassis
TA-11	Cockpit/display bench	Displays, audio, controls, Android/HMI automation	Mode request, telltales, warnings, boot/restart and fallback indication	Cockpit
TA-12	Thermal chamber	Hot/cold chamber + full ECU workload	Compute derating, PMIC/memory/network margin, sensor calibration and recovery	Thermal/HW
TA-13	EMC chamber	CISPR 25 + ISO 11452 component setup	Conducted/radiated emissions and immunity at representative maximum load	EMC
TA-14	SI/PI lab	VNA/TDR/high-bandwidth scope/probes	LPDDR, high-speed SerDes/Ethernet, power rails and clock integrity	SI/PI
TA-15	Cybersecurity lab	Pen-test host, diagnostic/update tool, fuzzing and secure-key fixtures	Secure boot, debug, diagnostics, SecOC, gateway, OTA and compromised-IVI attacks	Cyber
TA-16	OTA/update rig	Multi-ECU update orchestration + power/network fault insertion	Dependency manifest, interrupted download/install, rollback and mixed-version recovery	Platform/Cyber
TA-17	EOL fixture	Production connector/loads/network + secure provisioning	Power/current, I/O/network, firmware/calibration/key provisioning, genealogy	Manufacturing
TA-18	Service tool environment	UDS/DoIP tester + authenticated service workflow	DTCs, routines, replacement/configuration, permissions and audit	Diagnostics/Service
TA-19	Vehicle validation fleet	Production-intent vehicles + logging	Road robustness, driver modes, thermal, diagnostics, post-SOP telemetry	Validation
TA-20	FRACAS/field analytics	Issue database + event/DTC/software/configuration correlation	Field failure closure, recurring issue detection, supplier corrective action	Quality/Program

15. Rev A2 Technical Audit Disposition and Design-Freeze Decisions

Rev A2 converts the external audit into controlled engineering decisions. The audit is treated as design input, not specification authority: valid concerns are adopted, illustrative numerical values are qualified as program targets, and recommendations that would weaken safety or create false compliance claims are corrected before allocation.

ID	Critique / Gap	Disposition	Rev A2 Decision
RV-A2-001	Power budget 75 W peak	Adopt with architecture refinement	Increase combined central ECU design envelope to 70 W nominal / 130 W sustained peak reference, with 150 W short electrical-front-end allocation. External display backlights/audio amplifiers remain separate.
RV-A2-002	Entire-ECU 110 W hold-up capacitor	Modify	Do not hold up the full HPA/cockpit domain. Cold crank is handled by a buck-boost front end; only the independent safety domain is energy-backed. 20 W for 100 ms from 12->6 V at 90% gives ~41 mF ideal before ESR/aging margin.
RV-A2-003	ASIL C torque arbitration vs ASIL D hazard	Adopt / resolve contradiction	SG-01 becomes ASIL D target. Authoritative torque arbitration/monitor/inhibit path is ASIL D or formally decomposed per ISO 26262-9. HPA torque/mode request remains untrusted QM/ASIL-supporting input.
RV-A2-004	Remove SecOC from 10 ms torque frames	Reject blanket rule / refine	AUTOSAR E2E is mandatory for safety integrity; SecOC is applied wherever TARA requires authenticity, including a 10 ms torque PDU if attack exposure warrants it. HSM acceleration and TARA-selected authenticator/freshness lengths must close latency and bus load.
RV-A2-005	Driver-mode jerk matrix	Adopt as calibration starting point	Keep values as program calibration targets, not universal safety limits. Add eligibility, boost timer, rollback, mode persistence and Track-mode legality gates.
RV-A2-006	50 ms mode transition	Clarify	50 ms applies to request qualification/acknowledgement. Actual torque/regen/chassis maps blend over 200-500 ms reference or faster only when safety intervention requires.
RV-A2-007	R155/R156/ISO24089	Adopt	Add type-approval / lifecycle compliance matrix and software-update governance. Apply current series/amendments at homologation freeze.
RV-A2-008	Flight recorder 30 s	Adopt and extend	Use RAM circular capture with trigger-based crash-safe NVM flush; 30 s pre + 30 s post reference, synchronized with current IEEE 802.1AS baseline where Ethernet time is used.
RV-A2-009	FFI via MPU/IOMMU	Adopt and extend	Add DMA containment, interrupt ownership, DDR/cache/memory-bandwidth QoS, separate reset/clock/power roots and local guest restart containment.
RV-A2-010	30% CPU headroom	Adopt as design target	Safety MCU >=30% WCET/CPU margin at worst-case interrupt/network load; HPA critical-services reserve separately budgeted. Margin is measured, not inferred.
RV-A2-011	110-130 W implies thermal hierarchy	Adopt	Liquid/conduction cooling reference; shed GPU/NPU/Android background work before vehicle-control degradation. Safety MCU remains thermally/power independent.
RV-A2-012	Compliance matrix incomplete	Expand	Add ISO 16750-2:2023, ISO 11452-1:2025, CISPR25:2021, ISO11898-1:2024, ISO13400-2:2025, ISO6469-3:2021, UNECE R10/R100/R13-H/R79 interfaces, plus standards watchlist.
RV-A2-013	Hypervisor safety ambiguity	Close	Ordinary HPA hypervisor is QM unless safety-certified and used per safety manual. Safety-critical motion authority remains outside HPA. Safety display functions require safety-qualified path or independent fallback telltale.
RV-A2-014	Boot sequencing missing	New gap	Safety MCU, CAN and HV/torque supervision become ready independently of Android boot. Reference safety-domain

ID	Critique / Gap	Disposition	Rev A2 Decision
			ready <=150 ms; full cockpit services may boot later without blocking safe vehicle state.
RV-A2-015	Safe state is always torque zero	New gap / correct	Safe response is context dependent: immediate torque inhibit for unintended-positive-torque faults; controlled torque ramp/limited limp mode for other faults when the vehicle safety concept permits. Local inverter/ESC protection retains authority.
RV-A2-016	Track mode geofence as safety mechanism	Correct	Geofence may support UX/policy but is not relied upon as the only safety/legal control. Use market coding, explicit driver confirmation, vehicle-state prerequisites and ESC/local hard limits.
RV-A2-017	Dynamic memory / shared-resource starvation	New gap	Any safety-relevant HPA service must use bounded memory/resource quotas; Android/QM workloads can be dynamic but are rate/CPU/memory/GPU constrained so they cannot starve safety support or networking.
RV-A2-018	Safety telltale/cockpit dependency	New gap	Legal/safety-critical telltales and warnings require bounded availability and an independent source-of-truth. If HPA display domain is QM, provide safety overlay/fallback indicator or qualified safety rendering path.
RV-A2-019	OTA supply-chain completeness	New gap	Add SBOM, signed dependency manifest, vulnerability handling, crypto agility, certificate/key rotation, recovery and update-state interlocks.
RV-A2-020	Data privacy / driver profiles	New gap	Driver profiles, geofence/valet, cloud telemetry and event data require market privacy governance, purpose limitation, retention rules and access controls.
RV-A2-021	Charging/drive interlock authority	New gap	Add plug/charge-state/park/gear/HV READY interlocks so drive torque cannot be enabled during incompatible charging/service states.
RV-A2-022	Common-cause failure central ECU	New gap	One housing can still fail as one physical location. Add thermal, water ingress, power, oscillator/reset, connector and EMI common-cause analysis; preserve local inverter/ESC/BMS safety layers.

Release rule. Any audit observation that changes torque authority, ASIL allocation, power dissipation, network security, driver-mode behavior, homologation, or update strategy must be propagated through SYS.1, SYS.2, SYS.3, HSR/SWSR, DFMEA/TARA, DVP and RTM before design freeze.

16. Revised Central Compute Power, Thermal and Brownout Architecture

The original 75 W peak assumption is not retained for a premium centralized cockpit/VCU architecture. Rev A2 separates normal compute power, short electrical input capability and the safety-domain survival requirement so that the power tree can be designed without pretending the full Android/HPA domain must ride through every low-voltage disturbance.

Load Domain	Nominal W	Peak W	Design Rule	Qualification
Safety MCU + PMIC + safety CAN	12	20	Independent rail; torque/HV supervision survives HPA reset	ASIL-D safety path reference
HPA SoC CPU/GPU/NPU	35	75	Depends on nominated silicon and display/AI load	Throttle/shedding first
LPDDR5X memory	8	14	Multi-channel memory + ECC where supported	Budget bandwidth and thermal separately
UFS/NVMe storage	2	5	Update/log burst	No safety authority
Ethernet switch/PHYs	5	10	1G baseline; Multi-G future branch	Safety traffic isolated
Display/SerDes interface silicon	3	8	Pixel links excluded from vehicle Ethernet backbone	External panels excluded
Misc I/O / sensors / housekeeping	5	8	ADC/discretes/wake/secure elements	Includes engineering margin
Central ECU total	70	130	Sustained design target; 150 W short input allocation	External backlights/audio amplifier excluded

Controlled reference: 70 W nominal, 130 W sustained central-ECU peak and 150 W short electrical input capability. External display backlights, speaker/amplifier loads and externally powered antennas/sensors remain separate unless explicitly reallocated.

Cold-crank architecture: the ECU front end uses a buck-boost or equivalent automotive pre-regulator to support the released OEM/ISO electrical profile. Energy storage is credited only to the independent safety domain. For the Rev A2 design reference of 20 W for 100 ms from a 12 V storage rail down to 6 V at 90% efficiency, the ideal capacitance is about 41 mF; with a 1.5 design margin the starting allocation is about 62 mF. The full 130 W HPA domain is allowed to reset in a controlled manner.

Power Design Item	Rev A2 Baseline	Closure Evidence
LV front end	Buck-boost / wide-input architecture; 150 W short envelope	WCCA, cold crank, load dump/reverse battery, thermal and conducted-transient DVP
Safety rail	Independent PMIC/reset/watchdog/CAN supply; 20 W / 100 ms hold-up reference	Energy calculation, ESR/aging/cold margin, brownout fault injection
HPA rail	130 W sustained peak design reference; controlled reset permitted under severe LV fault	Load-step, PI, guest restart and recovery test
Thermal path	Conduction/liquid-assisted enclosure branch as required; QM compute sheds first	CFD, chamber, hotspot and throttling-state validation

17. Functional Safety Architecture - ASIL-D Torque Path and Context-Dependent Safe State

Rev A2 resolves the earlier integrity mismatch explicitly. SG-01 for unintended/excess positive torque is an ASIL-D target. The high-performance cockpit SoC is not an authoritative torque source; the independent safety MCU owns torque arbitration, monitoring and inhibit, while the traction inverter retains local torque/current/short-circuit protection.

Safety Function	Target	Architecture	Boundary
SG-01 Unintended positive/excess torque	ASIL D target	Safety MCU authoritative torque arbiter + independent torque monitor + hard inhibit; inverter local torque/SC protection remains independent	No HPA direct actuator authority
Nominal torque function	ASIL D or decomposed implementation	Primary command and independent monitor can use ISO26262-9 decomposition only after dependent-failure analysis	Decomposition is not automatic
Pedal acquisition	ASIL D path	Dual independent pedal channels, range/rationality/rate plausibility, brake override	Fault cannot increase torque
Delivered torque plausibility	ASIL D path	Compare command vs inverter actual torque/current/speed and acceleration plausibility	Mismatch -> inhibit/controlled degradation
HV READY authorization	ASIL C/D by item	VCU orchestrates; BMS owns pack contactors/precharge; inverter enable depends on valid HV state	No VCU bypass of BMS protection
Safety reaction	<=10 ms internal capability	DTI + qualification + inhibit-request budget; vehicle FTTI remains HARA/TSC-derived	Not a mechanical torque decay guarantee

The architecture does not assume that every fault should produce immediate zero torque. Immediate positive-torque inhibit is correct for unintended-acceleration faults, but some loss-of-function faults can require a bounded torque ramp or limited limp-home strategy if the vehicle FSC/TSC shows that this produces the lower residual risk.

Fault / Condition	VCU Response	Timing	Independent Layer	Recovery
Unintended positive torque / command corruption	Immediate positive-torque inhibit; brake/ESC retain authority	<=10 ms internal inhibit request	Inverter hard limiter + local protection	No limp torque until fault cleared
Pedal plausibility failure	Zero driver-demand torque; controlled propulsion fallback if vehicle-level safety concept allows	<=10 ms qualification/inhibit	Brake override active	Driver warning
Loss of one CAN safety bus	Continue on healthy redundant path if integrity valid	Within message timeout	No authority increase	Service DTC
Loss of both torque communication paths	Controlled torque removal / local inverter safe state	TSC-defined	Local inverter watchdog	Context-dependent coast/stop
HPA/Android crash	Safety MCU continues torque/HV supervision; cockpit degraded	No motion interruption required	Independent safety power/reset	Restart HPA guest
Thermal overload HPA	Shed GPU/NPU/background, then controlled HPA shutdown	Seconds-level thermal state	Safety MCU unaffected	Preserve legal/safety warning path
BMS SOP invalid	Reduce torque/regen to conservative limits or zero as allocated	<=SOP timeout	No increase in torque/charge	Driver warning
ESC/ABS active	Revoke/reduce regen and propulsion as requested by ESC	<=10 ms control loop	Brake ECU final authority	No driver-mode override
Crash input valid	Torque zero + HV de-energization request per vehicle safety concept	Vehicle crash FTTI	BMS/inverter local crash paths remain	No software re-enable

18. Platform Virtualization, Freedom-from-Interference and Resource Budgets

A centralized ECU can be production worthy only if physical consolidation does not create unrestricted cross-domain failure propagation. Rev A2 adds resource-level FFI requirements for CPU, memory bandwidth, DMA, interrupt ownership, power/reset and clocking.

Resource	Rev A2 Budget	Design Rule	Verification
Safety MCU CPU	<=70% worst-case utilization	Leave >=30% WCET/interrupt margin at max CAN/fault load	WCET + stress HIL
Safety MCU RAM/Flash	<=75% allocated	Keep diagnostic/update margin; no runtime heap in critical loops	Link map + runtime monitor
HPA CPU/GPU/NPU	<=75% sustained released workload	Reserve for UI spikes/update/background isolation	Target workload stress
HPA memory bandwidth	<=65% sustained released workload	Reserve >=35% to avoid DDR contention under multi-display/network bursts	SoC performance counters
HPA RAM	<=80% normal high-watermark	Bound each VM/cgroup; safety-support services preallocate bounded pools	Long-duration stress
DMA / IOMMU	Explicit allow-list	QM guest DMA cannot reach safety memory/MMIO	IOMMU fault injection
Cache / interconnect QoS	Priority/way/bandwidth partitioning where silicon supports	QM workload cannot starve safety-support/control IPC	Contention benchmark
Interrupt ownership	Static routing	Safety interrupts cannot be masked by Android guest	Fault injection
Power / reset	Independent roots	HPA reset/brownout does not reset safety MCU/PMIC/CAN	Brownout/reset test
Clock	Independent safety clock supervision	HPA PLL/clock fault cannot remove safety monitor timing	Clock fault injection

Hypervisor boundary. In the reference architecture the HPA hypervisor and Android/Linux guests remain non-authoritative for motion control. Any future decision to host a credited safety function on the HPA must nominate a safety-qualified hypervisor/OS combination, follow its safety manual, repeat dependent-failure analysis, and re-open the safety case.

Boot State	Function	Reference Timing	Safety Rule
OFF	Safety MCU off; always-on wake monitor active	Sleep current OEM target	No torque/HV authority
WAKE	Safety PMIC/MCU powers first; self-test/watchdog/HSM init	<=50 ms start reference	HPA still off
SAFETY_READY	CAN A/B, pedal/discrete, torque monitor and HV-state supervisor available	<=150 ms reference from valid wake	Android/cockpit boot not required
HPA_BOOT	SoC/hypervisor/guests start under independent power/reset supervision	<=2 s service target, platform-specific	Safety functions continue
DRIVE_READY	Valid BMS/ESC/inverter/HV state + no critical DTC + safety checks complete	Vehicle-state dependent	Mode manager enabled
HPA_RESTART	Local guest/SoC reset after failure	Safety domain remains active	Torque/HV authority retained by safety MCU
SHUTDOWN	Torque zero/HV state safe; logs flush; HPA off before safety domain if possible	Controlled	Safety MCU last-off

19. Multi-Driver Mode Rev A2 - Eligibility, Blending, Overboost and Rollback

The driver-mode function is treated as a bounded calibration service, not as a direct source of torque authority. The user/HMI requests a mode; the safety domain validates eligibility and then applies a jerk-bounded transition. This avoids both the original 50 ms ambiguity and the risk of Track-mode policy becoming a safety bypass.

Mode	Vehicle State	Eligibility	Power Cap	Regen	Jerk Target	Release Boundary
Eco	Any drive-ready	SOP valid; no safety inhibit	110 kW reference or lower	High regen	<=1.2 m/s3 design target	User selectable
Normal	Any drive-ready	SOP valid	Up to 220 kW	Medium	<=2.0 m/s3 target	Default after fault/update unless OEM policy differs
Sport	Forward / suitable speed	BMS/inverter thermal/SOP margin; ESC ready	Up to 220 kW	Low-medium	<=3.5 m/s3 target	Reject on thermal/traction degradation
Track	Explicit eligible state	Market/variant permit, explicit driver confirmation, ESC/local ECU support, tyre/chassis state, no critical DTC	Supplier-approved boost only	Configurable	<=5.0 m/s3 track calibration target	Geofence optional support, not sole safety/legal control
Valet	Drive-ready	Signed owner/fleet policy	80 kW default or lower	Standard	<=1.5 m/s3 target	Cannot raise cap without authorization
Mode Transition	Any accepted request	Safety MCU validates speed/gear/SOP/thermal/brake/ESC	Cross-fade 200-500 ms	Cross-fade	Jerk bounded by active mode/vehicle stability	Ack <=50 ms separate from map blend

- Accepted/rejected mode acknowledgement target: <=50 ms. This is not the torque-map cross-fade duration.
- Normal mode-map/regen/chassis transition reference: 200-500 ms with jerk supervision; emergency safety intervention can be faster.
- Track mode requires market/variant enable, explicit driver confirmation and local ESC/ABS/chassis acceptance. Geofence can support policy but is not the only safety or legal gate.
- Time-limited boost uses a persistent thermal/energy-debt accumulator so rapid Sport/Track cycling cannot reset the protection budget.

20. Global Compliance, Type-Approval and Standards Watchlist

The Rev A2 compliance architecture distinguishes engineering standards from vehicle type-approval regulations and from OEM-specific vehicle requirements. Applicable regulation series and amendment levels are frozen at the homologation gate rather than assumed from a static document date.

Domain	Standard / Regulation	2026 Status / Note	VCU220 Evidence
Functional safety	ISO 26262:2018	Published baseline; third edition under development in 2026	Safety lifecycle/HARA/FSC/TSC/HW/SW safety case
Cybersecurity	ISO/SAE 21434:2021	Published	TARA, lifecycle cyber risk management
Software update	ISO 24089:2023	Published	Update engineering, dependency/rollback evidence
ASPICE	Automotive SPICE 4.0	Program process baseline	SYS.1/SYS.2/SYS.3 + HWE/SWE/SYS.4/5 handoff
Electrical loads	ISO 16750-2:2023	Current published edition	LV operating/transient test profile plus OEM requirements
Conducted transients	ISO 7637-2:2011	Published/confirmed; 12/24V	Bench transient immunity; OEM pulse profile still controls
EMC immunity	ISO 11452-1:2025 + relevant parts	Current general principles edition	Component immunity plan
Emissions	CISPR 25:2021	Published	On-board receiver protection / component emissions
CAN	ISO 11898-1:2024	Published; includes CAN Classic/FD/XL framework	Network architecture/future branch
DoIP	ISO 13400-2:2025	Current published transport/network edition	Diagnostics over IP
UDS	ISO 14229-1:2020	Published	Diagnostic services
EV electrical safety	ISO 6469-3:2021	Published; revision underway	HV state/interface vehicle safety evidence
Ethernet timing	IEEE 802.1AS-2025	Active; supersedes 2020	gPTP/time-sync architecture
Ethernet sleep/wake	OPEN Alliance TC10	Automotive Ethernet sleep/wake specifications	Partial networking/wake design
Cyber type approval	UN R155	Apply current series/amendments at market homologation	CSMS/type approval evidence
Software update type approval	UN R156	Apply current series/amendments at market homologation	SUMS/RXSWIN/update compliance
Vehicle EMC	UN R10	Apply applicable current series at homologation	Vehicle/component EMC type approval
EV powertrain safety	UN R100	Vehicle-level applicability	Rechargeable energy storage/electric powertrain interfaces
Braking interface	UN R13-H / applicable market braking regulation	Vehicle-level interface	Regen/friction blend must not compromise braking compliance
Steering interface	UN R79 / applicable market steering regulation	Vehicle-level interface	Driver-mode steering request remains inside EPS/regulatory boundary

20.1 Standards watchlist

Item	Status	Program Action	Owner
ISO 26262 Ed.3	Under development in 2026	Run delta review before SOP if edition changes before compliance freeze	Safety Lead
UN R155/R156 amendments	Active interpretation/amendment activity in 2025-2026	Freeze applicable regulation series at homologation gate	Cyber/Homologation
UN R10 07-series evolution	Current series/amendments evolving	Use market-specific applicable series at EMC homologation freeze	EMC/Homologation
AUTOSAR R25-11	Current release in 2026; program may remain on controlled R24-11 baseline	Perform delta assessment; do not upgrade stack mid-program without impact analysis	SW Architect
IEEE 802.1AS-2025	Supersedes 802.1AS-2020	Use 2025 baseline for new design unless supplier stack is frozen older with compatibility evidence	Network Architect
ISO 6469-3 next edition	Under development	Track changes affecting HV operational/interface safety	HV Safety

21. Cybersecurity, OTA, Software Supply Chain and Privacy Governance

Cybersecurity is integrated with the safety architecture but is not allowed to degrade deterministic motion-control timing. AUTOSAR E2E and SecOC are treated as separate mechanisms: E2E addresses communication safety, while SecOC is applied according to the TARA where message authenticity and freshness are required.

Function	Requirement	Implementation	Boundary
Secure boot	HSM roots on Safety MCU and HPA; measured/verified chain	Anti-rollback, signed boot/config	No unsigned recovery image
OTA	A/B or equivalent atomic strategy with signed dependency manifest	Update only in allowed vehicle state; power-loss recovery	ISO24089 / R156 process
SBOM	Machine-readable ECU software/component inventory	Link version -> vulnerability status -> affected VIN/variant	Supply-chain governance
Crypto agility	Algorithm/key-length abstraction and certificate rotation	Avoid hardcoded lifecycle crypto dead-end	OEM policy controls algorithms
Debug	Production debug disabled or authenticated	Physical service policy and tamper logging	No generic unlock
SecOC	TARA-driven on safety/security relevant PDUs	HSM acceleration; schedule verified	Independent from E2E
Ethernet	VLAN/ACL/rate limiting; MACsec where threat analysis allocates	HPA/IVI not permitted direct safety-network injection	Gateway default deny
IDS	CAN/Ethernet anomaly monitoring without blocking safety timing	Event forwarding to backend when available	Safety reaction takes priority

21.1 Privacy and driver-profile data

Data Domain	Principle	Control	Boundary
Driver profile / mode preference	Store minimum attributes required for feature	Encrypted at rest; owner/profile access controls	Market privacy/consent rules
Valet/geofence policy	Treat location and owner constraints as personal/security data	Purpose limitation and retention policy	Geofence not sole safety control
Flight recorder	Separate safety/event channels from entertainment history	Role-based service access; retention schedule	Legal/insurance rules vary
Cloud telemetry	Opt-in/contractual lawful basis where required	Data minimization, pseudonymization, backend ACL	No cloud dependency for safety
Voice/assistant/cockpit analytics	Outside safety authority	Dedicated privacy consent and delete/export workflow	No access to raw torque controls

22. Rev A2 Program Gates, Evidence Closure and Production-Intent Readiness

The hardening work changes both architecture and program execution. The program plan therefore adds explicit architecture closure gates before Sample B/C so that ASIL-D torque independence, HPA power/thermal capability, network security schedulability, critical telltales, OTA recovery and product lifecycle support cannot remain late integration findings.

Gate	Mandatory Rev A2 Evidence	Release Condition
Architecture CDR	Safety MCU/HPA partition, 130 W thermal feasibility, CAN schedule, safe-state matrix, driver-mode eligibility	No open critical architecture decision
Sample A exit	Safety boot independent of Android, PMIC/load-step, basic torque monitor/inhibit, HPA resource isolation	Bench evidence traceable to SYS.3
Sample B exit	Full HIL torque/regen/mode faults, E2E/SecOC scheduling, OTA recovery, HMI fallback, thermal shedding	No unresolved ASIL-D or cyber-critical defect
DV exit	EMC/electrical/environmental, dyno/vehicle dynamics, penetration testing, update robustness and fleet recorder	DVP evidence complete for released variant
PV/SOP	EOL isolation/configuration screen, homologation evidence, SUMS/CSMS artifacts, service/lifecycle plan	Production release board approval

Rev A2 Controlled Count	Value
RA	68
SYS.1	58
SYS.2	115
HARA	48
SYS.3	246
HSR	70
SW Safety	72
DFMEA	120
TARA	40
ICD	38
DVP	492
Validation	50
RTM	246

23. 400 V / 800 V Product-Line Variant and Reuse Architecture

The VCU220 documentation is intended to remain stable across 400 V and 800 V propulsion variants. The software/safety architecture therefore consumes normalized inverter/BMS capability interfaces instead of embedding pack-voltage assumptions into torque control.

Design Area	400 V Variant	800 V Variant	Common VCU Rule
Peak propulsion	220 kW reference; ~579 A battery-path plausibility at 95%	220 kW reference; ~289 A plausibility at 95%	VCU commands torque/power limits, not raw HV current
BMS / inverter ICD	Variant-specific V/I/SOP/precharge envelope	Variant-specific V/I/SOP/precharge envelope	Signed variant binding and range plausibility
HV state	BMS-owned contactor/precharge reference	BMS-owned contactor/precharge reference	Same VCU HV state machine; separate calibration
Thermal	Variant component capability	Variant component capability	Normalized battery/inverter/motor capability percentages
DVP	400 V plant/inverter/BMS models	800 V plant/inverter/BMS models	Common test intent, variant data sets

24. Production Readiness, EOL Quality and Common-Cause Control Plan

Production intent is not reached only by completing software tests. Rev A2 makes physical centralization risks and EOL partition-isolation checks explicit program deliverables.

Production Characteristic	Control Method	Evidence / Gate
Safety rail independence	Separate PMIC/reset supervision; HPA reset injection	EOL/FAT + DV brownout report
Torque inhibit path	I/O/readback and inverter acknowledgement	100% functional screen / controlled test coverage
Secure provisioning	HSM keys/certificates/variant/calibration signature	Station audit + genealogy
Mode calibration binding	VIN/powertrain/market compatibility	Wrong-variant negative test
HPA partition isolation	Guest reset, malformed IPC/network, IOMMU configuration	Validated EOL control plan and DV fault injection
Thermal interface	TIM/heatsink/contact pressure/temperature monitor	Process capability + thermal end-of-line sample audit
Network identity	CAN/Ethernet IDs/MAC/VLAN/security profile	Topology scan and signed configuration
Common-cause controls	Enclosure ingress, power connector, clock/reset, EMC roots	CCF review + environmental/EMC DVP

25. Rev A2 Open-Item Register and Final Freeze Ownership

A reusable technical design must distinguish deliberate open program inputs from missing engineering. The following items remain controlled freeze inputs rather than hidden assumptions.

Open Input	Owner	Freeze Gate	Mandatory Closure Evidence
Final safety MCU / HPA SoC nomination	EE/System Architect	Architecture CDR	Target benchmark, safety manual, power/thermal and lifecycle evidence
Vehicle HARA/FSC/TSC and final ASIL allocation	Functional Safety	FSC/TSC release	Item definition, HARA, dependent-failure analysis and safety case
OEM LV transient profile	EE / Vehicle Integration	HW design freeze	ISO/OEM electrical profile and bench test plan
CAN/Ethernet traffic matrix	Network Architect	ICD freeze	CANoe schedulability and Ethernet QoS/TSN evidence
Critical telltale implementation	Cockpit/FuSa	Cockpit architecture freeze	Safety display/fallback concept and HMI fault tests
Track/Sport market availability	Homologation/Vehicle Dynamics	PV configuration freeze	Regulatory/legal review, ESC/chassis validation and signed coding
Cyber/OTA production profile	Cyber/SUMS	SOP software freeze	TARA, HSM/PKI, SBOM, R155/R156/SUMS evidence
Supplier lifecycle/PCN/EOL contracts	Program/Procurement	SOP sourcing freeze	15-year support/alternate/spares strategy